Coastal Aquaculture Pond Recognition via Scale and Boundary Dynamic Awareness Network in Remote Sensing Imagery

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***Abstract*—Coastal aquaculture ponds, vital to marine aquaculture, are widely distributed along coastlines and hold significant economic and ecological value. Accurately mapping their distribution and types is essential for optimizing marine spatial resources and protecting coastal ecosystems. While deep learning has improved detection from remote sensing imagery, precise recognition still faces challenges, such as large-scale and shape variations, blurred boundaries in dense areas, and a lack of high-resolution fine-grained datasets. To this end, a scale and boundary dynamic awareness network (BDA-Net) is proposed in this article. In particular, A Taylor-optimized scale-adaptive feature interaction (TSFI) structure is designed to achieve efficient multiscale feature adaptation for diverse pond sizes and shapes, overcoming fixed-scale Transformer limitations while preserving accuracy with lower computation via Taylor approximation. Moreover, a frequency-guided adaptive boundary convolution (FABC) is developed to resolve boundary confusion and detail loss by adaptively fusing multidirectional frequency features, enhancing contour-aligned sampling for improved fine-grained discrimination and recognition performance. In addition, a high-resolution fine-grained dataset is constructed to fill the gap in low spatial resolution and coarse category subdivision of existing aquaculture pond datasets. Extensive experiments on this dataset and public benchmarks demonstrate that the proposed BDA-Net achieves an intersection over union (mIoU) of 85.71%, outperforming the existing methods by over 5.45%.**

***Index Terms*—Aquaculture pond recognition, boundary-aware segmentation, fine-grained remote sensing dataset, multiscale adaptation.**

Received 27 September 2025; revised 2 December 2025; accepted 16 December 2025. Date of publication 19 December 2025; date of current version 2 January 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62501163, in part by Fujian Provincial Natural Science Foundation of China under Grant 2024J01277, in part by China Postdoctoral Science Foundation under Grant 2024M762192, and in part by the Foreign Cooperation Project of Fujian Province of China under Grant 2025I0008. *(Corresponding author: Hua Su.)*

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Digital Object Identifier 10.1109/TGRS.2025.3646224

I. Introduction

COASTAL aquaculture ponds, which form a vital part of marine aquaculture infrastructure, are extensively distributed along intricate coastlines and hold considerable economic and ecological value [1]. Accurate mapping of their spatial distribution and classification of their categories are crucial to optimizing marine spatial resource allocation and protecting the coastal ecological environment [2], [3].

Significant progress has been made in the extraction of aquaculture ponds recently [4], [5]. Advances in deep learning [6], [7], [8] have significantly promoted the automated and intelligent recognition of aquaculture ponds from remote sensing imagery [9], [10]. Numerous models have been proposed for this task, such as those derived from DeepLab series [11], [12], U-Net [13], [14], [15], and Transformer-based architectures [16], [17], [18], all demonstrating promising recognition performance. However, the advancement of such methods is constrained by the scarcity of high-resolution and finely annotated datasets. Most existing studies rely on medium- to low-resolution remote sensing images, which limits detailed analysis and still rely primarily on traditional methods with limited automation. Furthermore, fine-grained categorization of pond types has received limited attention, and many current models focus on extracting large aquaculture regions rather than precisely delineating individual pond boundaries. Therefore, there is a pressing need for an intelligent approach capable of high-precision extraction and fine-grained classification of aquaculture ponds from high-resolution remote sensing imagery to better support modern aquaculture management and development. In coastal areas, the distribution and morphology of aquaculture ponds show clear regional differences, as shown in Fig. 1(a). In coastal plains, these ponds are usually regular in shape and neatly arranged. However, in hilly areas, ponds are often constructed following the terrain. They extend along the coastline in winding distributions, with significant variations in shape and size, and exhibit irregular spatial layouts. This variability imposes high demands on the multiscale adaptation capability of recognition models.

Due to the locality of the convolutional layers, traditional CNN models are limited in capturing global contextual relationships. Consequently, they often struggle to effectively model long-range semantic dependencies and achieve holistic

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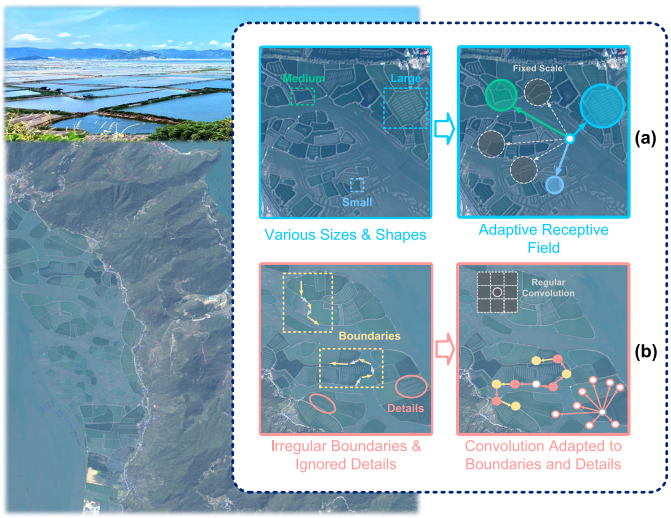


Fig. 1. Challenges faced in intelligent recognition of aquaculture ponds. (a) Fixed-scale models struggle with ponds of varying sizes and shapes. (b) Regular convolutions cannot extract irregular pond boundaries well and lose details.

scene understanding. In contrast, Transformer-based methods provide a new alternative for aquaculture pond extraction from remote sensing imagery by leveraging their strong global modeling capacity and have shown promising results in practical applications. Building on this, the Swin Transformer [19] introduces a shifted window mechanism, which reduces computational complexity while allowing cross-window interaction. This design effectively captures multiscale features and preserves global modeling capacity. Similarly, UNetFormer [20] combines the UNet architecture with the Transformer attention mechanism, enabling joint modeling of local and global information in the decoder and enhancing the interaction of global-local features. Nevertheless, aligning features across different scales remains a challenge in multiscale processing, particularly when dealing with significant variations in object sizes. Consequently, more sophisticated alignment mechanisms are often necessary to improve model performance.

Aquaculture ponds are typically constructed by excavating or reclaiming land following the natural terrain and are filled with seawater, as shown in Fig. 1(b). The materials used for the embankments, such as concrete or soil, contribute to diverse boundary morphologies and complex reflectance characteristics, making them easily confused with surrounding terrestrial areas. In densely distributed regions, the ponds exhibit curved and interconnected contours, leading to mutual interference of image features and making it difficult to distinguish individual ponds accurately. Most existing semantic segmentation methods [21] tend to coarsely extract entire aquaculture areas as a whole, lacking precise delineation of individual pond boundaries and internal structures [e.g., ponds equipped with aerators or containing cages or greenhouses, as shown in Fig. 1(b)]. As a result, they fall short in supporting practical applications that require high-precision boundary recognition and fine-grained management.

To address these challenges, dilated convolution [22] is widely used to enlarge the receptive field while preserving spatial resolution, yet its fixed kernel shape limits its ability to detect elongated and tubular structures like pond boundaries. Dynamic snake convolution (DSC) [23] offers a valuable reference by adaptively focusing on slender and curved local structures to better capture tubular features. Edge-aware methods [24], [25] have also been incorporated to enhance boundary segmentation. Furthermore, the subtle differences among various pond types pose significant challenges for fine-grained classification. For fine-grained classification, deep networks are employed to locate and align object parts [26], while category-aware attention mechanisms [27], [28], and generative classifiers [29] are introduced to strengthen feature discrimination and model underlying data distributions. Nevertheless, existing methods still struggle to accurately capture the nonlinear contour structures of pond embankments. Conventional convolution kernels exhibit limited capability in modeling highly curved geometric features. Moreover, during training, these approaches often overlook fine-grained details, resulting in insufficient intraclass differentiation and constraining performance in fine-grained classification tasks.

Based on the above analysis, current aquaculture pond recognition primarily faces two major challenges. On one hand, aquaculture ponds exhibit significant diversity in morphology and scale. The use of fixed receptive fields often leads to over-smoothing of small objects and fragmentation of large structures. On the other hand, the boundaries and fine-grained features of aquaculture ponds are often obscured by complex environmental interference and dense distribution, which hampers accurate distinction and limits overall recognition precision. Therefore, motivated by the above considerations, a scale and boundary dynamic awareness network (BDA-Net) is proposed in this work. The main contributions are summarized as follows.

1) A Taylor-optimized scale-adaptive feature interaction (TSFI) structure is proposed and utilized in BDA-Net's global-local decoder to address the significant morphological and scale variations in aquaculture ponds. It effectively overcomes the limitation of fixed-scale Transformer layers in flexibly adapting to varying object sizes. Furthermore, it employs the Taylor approximation to reduce computational overhead, improving accuracy while avoiding significant efficiency loss.
2) To tackle boundary confusion and detail loss in aquaculture ponds due to complex contours and environmental interference, a frequency-guided adaptive boundary convolution (FABC) is designed. It adaptively fuses multidirectional high- and low-frequency features aligned with contour gradients. This enhances the extraction of intricate boundaries and improves discrimination of fine-grained categories, leading to better intraclass feature alignment and superior recognition and classification performance.
3) A high-resolution fine-grained pond recognition (FGPR) dataset was constructed to address two key limitations of existing datasets: low spatial resolution and the lack of subdivided pond categories. Extensive experiments conducted on both the newly developed dataset and public benchmarks demonstrate the proposed method's

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strong performance in fine-grained recognition accuracy and cross-scenario generalization.

The remainder of this article is organized as follows. Section II reviews the related work. Section III provides a detailed explanation of the proposed BDA-Net. In Section IV, extensive experiments are conducted that demonstrate the performance of the proposed BDA-Net. Section V draws conclusions and prospects.

II. Related Works

A. Aquaculture Pond Recognition Based on Deep Learning

The object-oriented method serves as an important approach for extracting aquaculture ponds. Simulating human visual cognition and integrating multiple features of objects significantly improves classification accuracy and practical utility of results, demonstrating particular advantages in scenarios with complex land cover types and rich details. Chen et al. [30] proposed an automated extraction framework that deeply integrates evapotranspiration-related indices and machine learning to address the poor adaptability of segmentation thresholds when local models are generalized to a national scale. Ren et al. [31] combined an optimal segmentation scale with a random forest classifier to map coastal aquaculture ponds in China from 1984 to 2016. However, the object-oriented method involves complex processes and relies heavily on manual intervention and experience, resulting in limited automation and suboptimal accuracy. Deep learning advances in semantic segmentation offer enhanced accuracy and automation, enabling large-scale intelligent extraction and recognition of aquaculture ponds [9]. Zeng et al. [32] introduced a fully convolutional network with rowwise and columnwise self-attention to extract aquaculture ponds from high-resolution remote sensing imagery. For weak-signal environments, Liu et al. [11] applied DeepLabv3 with multisource features like NDWI and G/R ratio to identify and classify mariculture areas. U-Net-based models are widely adopted for aquaculture segmentation due to their strong performance [12]. Yu et al. [13] designed a composite framework using two UNet++ sub-networks to generate marker and foreground functions, integrated with a marker-controlled watershed algorithm for refined segmentation. Cui et al. [14] enhanced the U-Net with a PSE structure to extract coastal raft aquaculture from 1.8-m imagery, reducing adhesion effects. The U-Net architecture utilizes skip connections to merge shallow and deep features, recovering fine details.

Transformers have also been employed for global context in aquaculture mapping. Fu et al. [17] proposed TCNet, a hybrid Transformer-CNN model that combines local feature extraction and long-range dependence modeling to delineate mariculture zones. Su et al. [33] developed Raft-Former, which uses a global-local fusion module to integrate multiscale features and handle scale variation in raft aquaculture segmentation. Nevertheless, current models often overlook detail and local features, limiting species-level categorization, and fine-grained recognition of nearshore aquaculture ponds remains understudied.

B. Extraction of Multiscale Features

In remote sensing image segmentation, accurately capturing multiscale target features is crucial. Most current methods use convolutional neural networks (CNNs) and dilated convolutions to extract local features [22]. These approaches apply convolutional layers with different dilation rates to capture multiscale context. However, their fixed kernel shapes struggle with the irregular boundaries of nearshore aquaculture ponds, and discrete dilation rates cannot represent the continuous scale variations common in coastal zones. To tackle this, Qi et al. [23] introduced DSC, which uses deformable convolutions (DCN) to better capture thin and curved structures. CNNs are good at capturing spatial details, but due to the local nature of convolution, they often miss global semantic context.

In contrast, Transformers have become popular in semantic segmentation by effectively modeling global relationships [34], [35]. For example, Liu et al. [19] proposed Swin Transformer, which uses a shifted window attention mechanism to lower computation cost while improving representation power. Wang et al. [36] proposed a multiscale uncertainty consistency (MSUC) regularization method that constrains consistency among feature maps across different network layers. This approach enhances the multiscale learning capability of semi-supervised algorithms when processing unlabeled data. Xu et al. [37] combined Swin Transformer with ensemble learning for remote sensing image segmentation, alternating between local and global attention during training to reduce computational load. By the strengths of U-Net and Transformers, Wang et al. [20] designed UNetFormer, a U-Net-like Transformer with a global-local attention mechanism in the decoder. It shows better segmentation results than many lightweight models. Still, Transformers have limitations: their fixed receptive fields struggle with large-scale changes, often causing over-smoothing of local features and weak global structures. To improve multiscale attention, Ren et al. [38] proposed shunted self-a ttention, which allows each attention layer to mix information across scales. The key idea is to inject varied receptive fields into tokens. This helps self-attention learn relationships between objects of different sizes and improves multiscale capture. Nevertheless, existing scale-adaptive methods still lack efficient resource allocation. They cannot adjust computation based on local complexity, and high model complexity increases device burden.

III. Proposed BDA-Net Methodology

The proposed BDA-Net, illustrated in Fig. 2, is structured as follows. First, the model employs a U-shaped encoder–decoder architecture that integrates a CNN-based encoder [see Fig. 2(a)] and a CNN+Transformer decoder [see Fig. 2(d)] to achieve global-local feature complementarity. Second, the TSFI module [see Fig. 2(b)] is introduced to handle large variations in pond size and shape. It adaptively fuses multiscale features by combining the global modeling capacity of Transformers with the local extraction ability of convolutions, while incorporating the Taylor approximation for computational efficiency. Third, the FABC module [see Fig. 2(c)] is designed to address blurred boundaries and fine-grained features in complex environments. It utilizes multilevel

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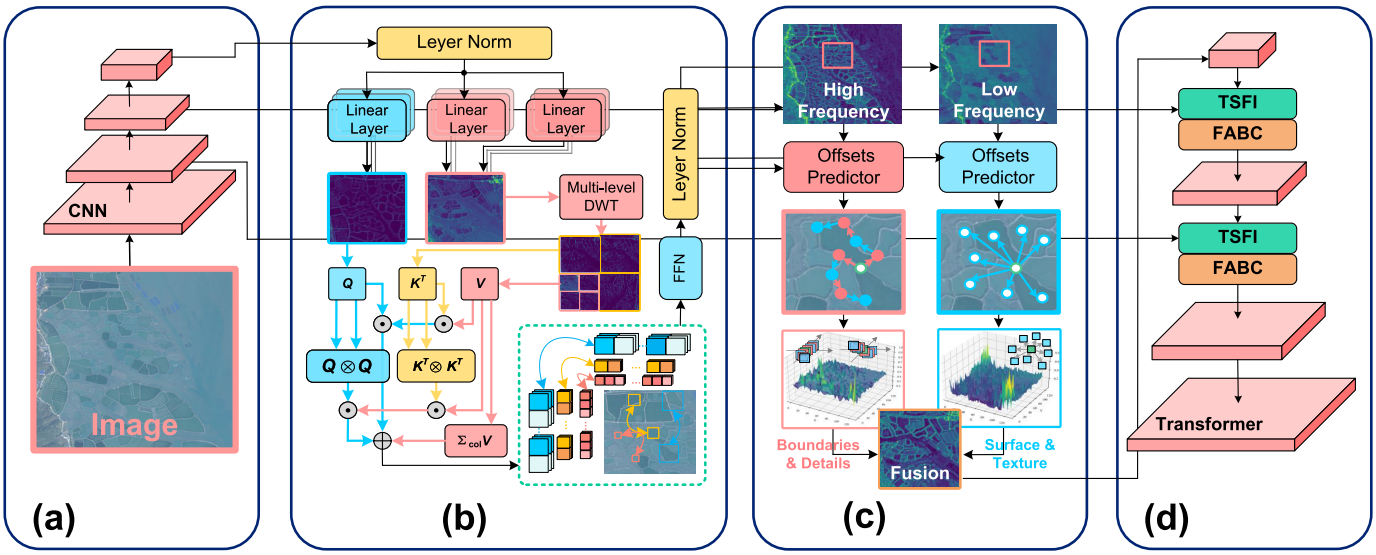


Fig. 2. BDA-Net framework comprises (a) the CNN encoder, (b) TSFI structure, (c) FABC, and (d) CNN+Transformer decoder.

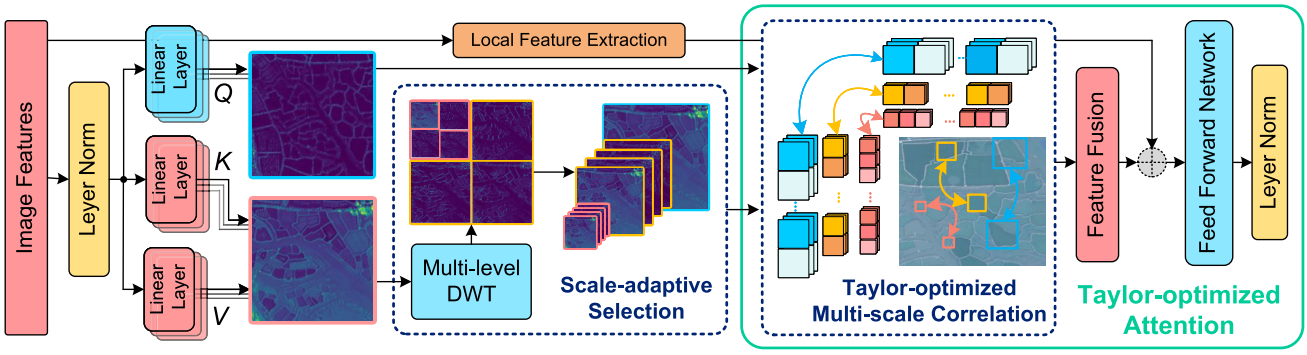


Fig. 3. Designed TSFI structure.

wavelet decomposition to fuse frequency features and align sampling with contours, significantly improving boundary precision and fine-grained classification.

A. Taylor-Optimized Scale-Adaptive Feature Interaction

Aquaculture ponds exhibit significant physical heterogeneity in real-world scenarios, manifested in a wide range of scales, irregular contours, and disordered spatial distribution. This unstructured, multiscale object requires models capable of cross-scale feature fusion, deformation-robust feature alignment, and context-aware spatial modeling.

A widely adopted strategy uses multiscale convolutional kernels to extract features across varying receptive fields [33]. However, the receptive field of a single convolutional layer remains both limited and localized. Convolutions with predefined kernel sizes are often inadequate in handling extreme scale variations, frequently resulting in missed detections of small ponds or loss of detail in larger ones. To overcome these limitations, the Transformer architecture, known for capturing global receptive fields, has been incorporated to enable the fusion of global and local features [20].

Define the input features as $\boldsymbol{X} \in \mathbb{R}^{H\times W\times C}$, where H, W, and C define the height, weight, and channel of feature tensors, respectively. After linear projection, we obtain the query, key, and value matrices

$$\boldsymbol{Q} = \boldsymbol{X}\boldsymbol{W}^{Q}, \quad \boldsymbol{K} = \boldsymbol{X}\boldsymbol{W}^{K}, \quad \boldsymbol{V} = \boldsymbol{X}\boldsymbol{W}^{V} \tag{1}$$

where $\boldsymbol{W}^{Q}$, $\boldsymbol{W}^{K}$, and $\boldsymbol{W}^{V} \in \mathbb{R}^{C\times d_k}$ are learnable weights. d_k is the dimension of each vector after projection. The basic self-attention calculation in the Transformer is presented as

$$\text{Attention}\,(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V}) = \text{Softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^{\top}}{\sqrt{d_k}}\right)\boldsymbol{V} \tag{2}$$

which requires $\mathcal{O}(n^2 d_k)$ operations due to the $n \times n$ ($n = H \times W$) matrix product $\boldsymbol{Q}\boldsymbol{K}^{\top}$. While this structure helps capture long-range dependencies, its scale is fixed for each Transformer unit. This makes it difficult to adapt to diverse object shapes and scales. Furthermore, the second-order computational complexity is computationally expensive for high-resolution remote sensing images.

Therefore, motivated by the above two considerations, the TSFI structure shown in Fig. 3 is designed to improve the scale of self-attention and avoid the huge computational overhead of multiscale calculations. First, we introduce a wavelet-based multiscale decomposition mechanism that enables dynamic scale adaptation within the attention. Reshape $\boldsymbol{K}$ and $\boldsymbol{V}$ to

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$H \times W \times d_k$ tensors, then apply 2-D discrete wavelet transform to each feature channel

$$\mathcal{W}(\boldsymbol{K}) = \left\{\boldsymbol{K}^{(LL)}, \boldsymbol{K}^{(LH)}, \boldsymbol{K}^{(HL)}, \boldsymbol{K}^{(HH)}\right\}$$
$$\mathcal{W}(\boldsymbol{V}) = \left\{\boldsymbol{V}^{(LL)}, \boldsymbol{V}^{(LH)}, \boldsymbol{V}^{(HL)}, \boldsymbol{V}^{(HH)}\right\} \tag{3}$$

where the superscripts denote different frequency sub-bands. Each $\boldsymbol{K}^{(l)}$ and $\boldsymbol{V}^{(l)}$ have dimension $n/4 \times d_k$. For the ith attention head and the lth scale of $\boldsymbol{K}$, i.e., $\boldsymbol{K}_i^{(l)}$, the adaptive weight is computed as

$$\alpha_{i,l} = \frac{\exp\left(\boldsymbol{z}_i^\top \boldsymbol{U}\boldsymbol{e}_l + b_l\right)}{\sum_{j=1}^{4} \exp\left(\boldsymbol{z}_i^\top \boldsymbol{U}\boldsymbol{e}_j + b_j\right)} \tag{4}$$

where $\boldsymbol{z}_i = \boldsymbol{W}_z \cdot \mathrm{GAP}(\boldsymbol{K}_i) \in \mathbb{R}^{d_z}$ is the global average pooling of $\boldsymbol{K}$, $\boldsymbol{e}_l \in \mathbb{R}^{d_z}$ is the scale embedding vector, $\boldsymbol{U} \in \mathbb{R}^{d_z \times d_z}$ is a learnable weight matrix, and $\boldsymbol{W}_z \in \mathbb{R}^{d_z \times d_k}$ is a projection matrix. The calculations for different scales of $\boldsymbol{V}$ are similar. This enables the Transformer to adaptively adjust features of different scales in different heads and layers according to learnable weights and model their multiscale nonlocal correlations.

However, multiscale representations also increase computational demands, which further compounds the complexity of the self-attention mechanism in the Transformer structure. To address this computational challenge, the multiscale attention mechanism is reformulated into a polynomial representation, whose complexity is then substantially reduced through a Taylor series approximation [39], [40], as illustrated in Fig. 4.

Due to the standard attention $\boldsymbol{A}^{(l)} = \mathrm{Softmax}(\boldsymbol{T}^{(l)})$ can be represented as

$$\mathrm{Softmax}\left(\boldsymbol{T}^{(l)}\right) = \mathrm{diag}^{-1}\left(\exp\left(\boldsymbol{T}^{(l)}\right)\mathbf{1}_N\right)\exp\left(\boldsymbol{T}^{(l)}\right) \tag{5}$$

where $\mathbf{1}_N$ is an N-dimensional column vector consisting entirely of ones, the softmax performs normalization through broadcasting. The nonlinear characteristic of softmax comes from the exponential part, while normalization is an independently handled linear operation, which can be aligned through subsequent adjustments [39], [40]. The Taylor expansion is a well-known mathematical tool for approximating exponential functions, and the expansion converges when the spectral norm of $\boldsymbol{T}^{(l)}$ is bounded. The error in the Taylor approximation will be discussed later. Therefore, we use the Taylor expansion to approximate the exponential part of the attention mechanism.

Thus, the scale-specific attention output for scale l is approximated as

$$\boldsymbol{A}^{(l)}\boldsymbol{V}^{(l)} \approx \left(\boldsymbol{I} + \boldsymbol{T}^{(l)} + \frac{\left(\boldsymbol{T}^{(l)}\right)^2}{2!} + \cdots + \frac{\left(\boldsymbol{T}^{(l)}\right)^k}{k!}\right)\boldsymbol{V}^{(l)} \tag{6}$$

where $\boldsymbol{T}^{(l)} = (\boldsymbol{Q}(\boldsymbol{K}^{(l)})^\top)/(d_k)^{1/2}$ donates the scale-specific similarity matrix. As illustrated in Fig. 4, typically truncated at second order ($k = 2$) for efficiency, thus,

$$\boldsymbol{A}^{(l)}\boldsymbol{V}^{(l)} \approx \left(\boldsymbol{I} + \boldsymbol{T}^{(l)} + \frac{\left(\boldsymbol{T}^{(l)}\right)^2}{2}\right)\boldsymbol{V}^{(l)}. \tag{7}$$

The zeroth-order term (identity mapping) is

$$\boldsymbol{A}_{(0)}^{(l)}\boldsymbol{V}^{(l)} = \boldsymbol{V}^{(l)}. \tag{8}$$

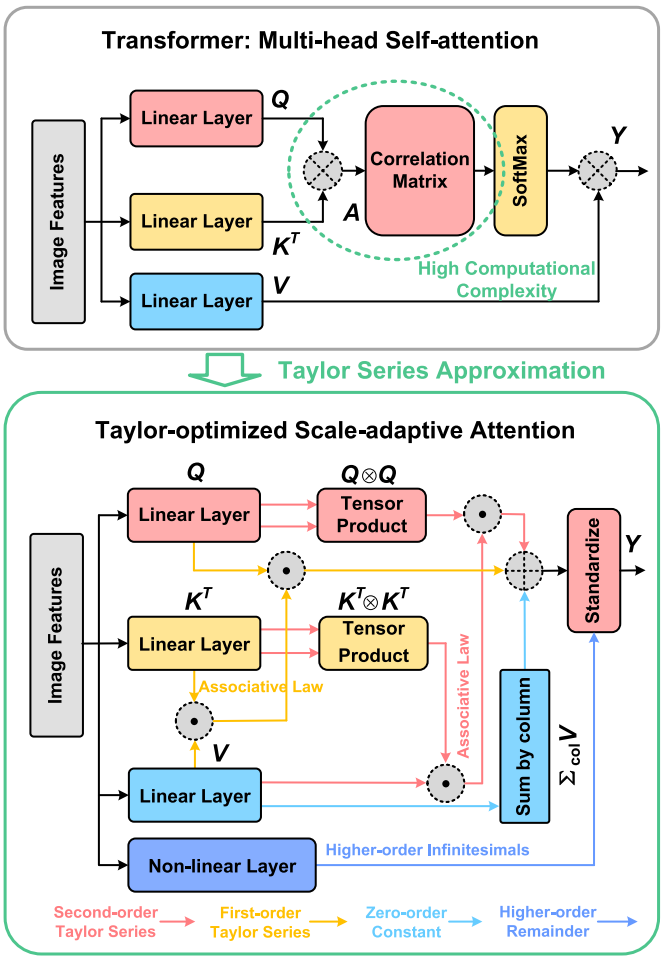


Fig. 4. Specific structure and calculation process of Taylor-optimized scale-adaptive attention.

The first-order term (linear term) is

$$\boldsymbol{A}_{(1)}^{(l)}\boldsymbol{V}^{(l)} = \boldsymbol{T}^{(l)}\boldsymbol{V}^{(l)} = \frac{1}{\sqrt{d_k}}\boldsymbol{Q}\left(\left(\boldsymbol{K}^{(l)}\right)^\top \boldsymbol{V}^{(l)}\right). \tag{9}$$

The second-order term (quadratic term) is

$$\begin{aligned}\boldsymbol{A}_{(2)}^{(l)}\boldsymbol{V}^{(l)} &= \frac{1}{2}\left(\boldsymbol{T}^{(l)}\right)^2\boldsymbol{V}^{(l)} \\ &= \frac{1}{2d_k}\boldsymbol{Q}\left(\left(\boldsymbol{K}^{(l)}\right)^\top\left(\boldsymbol{Q}\left(\left(\boldsymbol{K}^{(l)}\right)^\top \boldsymbol{V}^{(l)}\right)\right)\right).\end{aligned} \tag{10}$$

Furthermore, to alleviate performance loss caused by only the low-order approximation, a learnable projection matrix $\boldsymbol{P} \in \mathbb{R}^{d_k \times r}$ ($r \ll d_k$) is introduced to optimize the computation of higher order terms

$$\tilde{\boldsymbol{Q}} = \boldsymbol{Q}\boldsymbol{P}, \quad \tilde{\boldsymbol{K}}^{(l)} = \boldsymbol{K}^{(l)}\boldsymbol{P}. \tag{11}$$

Thus, the second-order term is approximated as

$$\boldsymbol{A}_{(2)}^{(l)}\boldsymbol{V}^{(l)} \approx \frac{1}{2d_k}\tilde{\boldsymbol{Q}}\left(\left(\tilde{\boldsymbol{K}}^{(l)}\right)^\top\left(\boldsymbol{Q}\left(\left(\boldsymbol{K}^{(l)}\right)^\top \boldsymbol{V}^{(l)}\right)\right)\right). \tag{12}$$

For higher order terms ($m \geq 2$), a recursive computation is used, that is,

$$\boldsymbol{A}_{(m)}^{(l)}\boldsymbol{V}^{(l)} \approx \frac{1}{m! \cdot d_k^{m/2}}\tilde{\boldsymbol{Q}}\left(\left(\tilde{\boldsymbol{K}}^{(l)}\right)^\top \boldsymbol{A}_{(m-1)}^{(l)}\boldsymbol{V}^{(l)}\right). \tag{13}$$

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Furthermore, we analyzed the complexity and approximation error of the proposed TSFI.

1) Complexity Analysis: The overall computational complexity is significantly reduced through Taylor approximation and low-rank projection [39], [40]. The dominant term of the self-attention mechanism is optimized from the original quadratic complexity $\mathcal{O}(n^2 d_k)$ to an asymptotically linear complexity $\mathcal{O}(L \cdot (nd_k^2 + k \cdot nd_k r))$, where $L = 4$ is the number of wavelet scales, k is the Taylor expansion order (typically set to 2), and r is the projection dimension ($r \ll d_k$). This is achieved while retaining the multiscale feature extraction capability, whose complexity $\mathcal{O}(\mathrm{nC})$ is linear and negligible. Furthermore, the memory requirements decrease substantially from $\mathcal{O}(n^2)$ to $\mathcal{O}(n(d_k + r))$.

2) Approximate Error Analysis: The Taylor approximation error is systematically decomposed into Taylor truncation error from finite-series expansion and low-rank projection error from dimensionality reduction [40]. The overall error bound satisfies

$$\left\| \boldsymbol{A}^{(l)} \boldsymbol{V}^{(l)} - \tilde{\boldsymbol{A}}^{(l)} \boldsymbol{V}^{(l)} \right\|_F \leq \epsilon \left\| \boldsymbol{V}^{(l)} \right\|_F \tag{14}$$

where $\epsilon > 0$ is the target error tolerance, governing approximation precision. The Taylor truncation error, arising from the kth order expansion of the exponential function, is bounded by the Taylor remainder theorem. The low-rank projection error [40], introduced by dimensionality reduction with projection dimension r, is controlled via the Johnson–Lindenstrauss lemma [41]. By selecting parameters k and r to constrain each error component to $\epsilon/2$, the total error remains within the specified tolerance. This provides theoretical guarantees for the Taylor approximation approach. Complete proofs are in the Appendix.

Thus, the output of the ith attention head is the weighted sum across all scales

$$\mathrm{Head}_i = \sum_{l=1}^{4} \alpha_{i,l} \sum_{m=0}^{k} \boldsymbol{A}_{(m)}^{(l)} \boldsymbol{V}^{(l)}. \tag{15}$$

The output of the multihead attention mechanism is

$$\mathrm{MHA}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = [\mathrm{Head}_1, \mathrm{Head}_2, \ldots, \mathrm{Head}_h] \boldsymbol{W}^O \tag{16}$$

where h is the number of attention heads, and $\boldsymbol{W}^O \in \mathbb{R}^{h \cdot d_k \times d_{\mathrm{model}}}$ is the output projection matrix. The specific calculation process of Taylor-optimized scale-adaptive attention is detailed in Algorithm 1. The complete global-local feature fused TSFI block is presented as

$$\boldsymbol{Z} = \mathrm{LayerNorm}(\boldsymbol{X} + \mathrm{MHA}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V})) \tag{17}$$

$$\boldsymbol{Y} = \mathrm{LayerNorm}(\boldsymbol{Z} + \mathrm{FFN}(\boldsymbol{Z})) + \mathrm{Conv}(\boldsymbol{X}) \tag{18}$$

where LayerNorm(·) denotes layer normalization, FFN(·) is the positionwise feed-forward network, and Conv(·) is the convolutional layers to extract local features.

The decoder of the proposed BDA-Net uses the TSFI structure to integrate features input locally through CNN lateral connections and global scale-adaptive features from deep Transformer layers refined via Taylor approximation. It effectively overcomes the limitation of traditional Transformers, which struggle to adapt flexibly to varying object sizes due to their fixed scale. Moreover, by leveraging Taylor series approximation, it reduces the computational complexity involved in multiscale processing, significantly improves accuracy, and avoids substantial efficiency loss.

Algorithm 1 Taylor-Optimized Scale-Adaptive Attention

Input: Input feature tensor $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, projection matrices $\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V \in \mathbb{R}^{C \times d_k}$, low-rank projection matrix $\mathbf{P} \in \mathbb{R}^{d_k \times r}$ ($r \ll d_k$), Taylor expansion order k (typically 2), scale embedding vectors $\{\mathbf{e}_l\}_{l=1}^4 \in \mathbb{R}^{d_z}$
Output: Final attention output $\mathbf{O}$

Step 1: Linear projection
$\mathbf{Q} \leftarrow \mathbf{X}\mathbf{W}^Q$; // Query projection
$\mathbf{K} \leftarrow \mathbf{X}\mathbf{W}^K$; // Key projection
$\mathbf{V} \leftarrow \mathbf{X}\mathbf{W}^V$; // Value projection
Step 2: Wavelet multiscale decomposition
$\mathcal{W}(\mathbf{K}) \leftarrow \{\mathbf{K}^{(LL)}, \mathbf{K}^{(LH)}, \mathbf{K}^{(HL)}, \mathbf{K}^{(HH)}\}$; // Decompose keys
$\mathcal{W}(\mathbf{V}) \leftarrow \{\mathbf{V}^{(LL)}, \mathbf{V}^{(LH)}, \mathbf{V}^{(HL)}, \mathbf{V}^{(HH)}\}$; // Decompose values
Step 3: Low-rank projection
$\tilde{\mathbf{Q}} \leftarrow \mathbf{Q}\mathbf{P}$; // Projected queries
for $l \leftarrow 1$ **to** 4 **do**
  $\tilde{\mathbf{K}}^{(l)} \leftarrow \mathbf{K}^{(l)}\mathbf{P}$; // Projected keys for scale l
end
Step 4: Compute attention weights
for $i \leftarrow 1$ **to** h **do**
  $\mathbf{z}_i \leftarrow \mathbf{W}_z \cdot \mathrm{GAP}(\mathbf{Q}_i)$; // Query representation
  for $l \leftarrow 1$ **to** 4 **do**
    $\alpha_{i,l} \leftarrow \frac{\exp(\mathbf{z}_i^\top \mathbf{U}\mathbf{e}_l + b_l)}{\sum_{j=1}^4 \exp(\mathbf{z}_i^\top \mathbf{U}\mathbf{e}_j + b_j)}$; // Scale weight
  end
end
Step 5: Compute attention outputs per scale
for $l \leftarrow 1$ **to** 4 **do**
  $\mathbf{A}_{(0)}^{(l)}\mathbf{V}^{(l)} \leftarrow \mathbf{V}^{(l)}$; // Zeroth-order term
  $\mathbf{A}_{(1)}^{(l)}\mathbf{V}^{(l)} \leftarrow \frac{1}{\sqrt{d_k}}\mathbf{Q}((\mathbf{K}^{(l)})^\top \mathbf{V}^{(l)})$; // First-order term
  for $m \leftarrow 2$ **to** k **do**
    $\mathbf{A}_{(m)}^{(l)}\mathbf{V}^{(l)} \leftarrow \frac{1}{m! \cdot d_k^{m/2}}\tilde{\mathbf{Q}}((\tilde{\mathbf{K}}^{(l)})^\top \mathbf{A}_{(m-1)}^{(l)}\mathbf{V}^{(l)})$; // Higher-order term
  end
  $\mathbf{Attn}^{(l)} \leftarrow \sum_{m=0}^{k} \mathbf{A}_{(m)}^{(l)}\mathbf{V}^{(l)}$; // Total attention for scale l
end
Step 6: Weighted fusion across scales
for $i \leftarrow 1$ **to** h **do**
  $\mathrm{Head}_i \leftarrow \mathbf{0}$; // Initialize head output
  for $l \leftarrow 1$ **to** 4 **do**
    $\mathrm{Head}_i \leftarrow \mathrm{Head}_i + \alpha_{i,l} \cdot \mathbf{Attn}^{(l)}$; // Add scaled attention
  end
end
Step 7: Combine heads
$\mathbf{O} \leftarrow \mathrm{Concat}(\mathrm{Head}_1, \ldots, \mathrm{Head}_h)\mathbf{W}^O$; // Output projection
return $\mathbf{O}$; // Final attention output

B. Frequency-Guided Adaptive Boundary Convolution

Aquaculture ponds are typically distributed in dense clusters along winding coastlines. Their banks, constructed from materials such as concrete or mud, often exhibit complex and irregular shapes, as illustrated in Fig. 5. In remote sensing imagery, these characteristics can cause ponds to be easily

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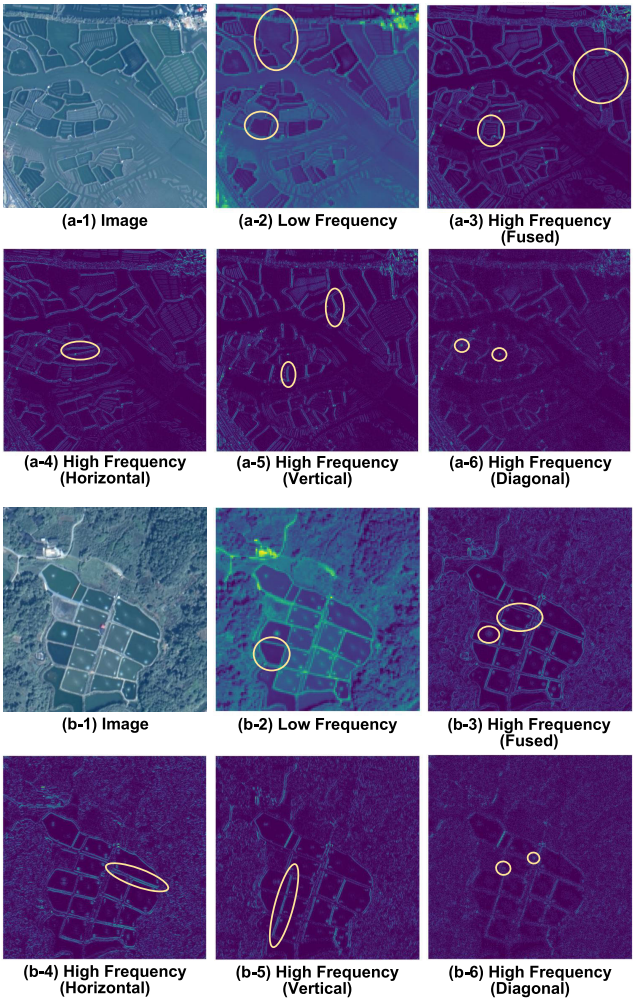


Fig. 5. Low-frequency and high-frequency features of aquaculture ponds. (a-1) and (b-1) original images, (a-2) and (b-2) low-frequency features obtained by wavelet decomposition, (a-3) and (b-3) three high-frequency feature components fused by wavelet decomposition, (a-4) and (b-4) high-frequency feature components in the horizontal direction, (a-5) and (b-5) are the high-frequency feature components in the vertical direction, and (a-6) and (b-6) high-frequency feature components in the diagonal direction. The yellow circles mark some key features obtained from each component.

confused with surrounding terrestrial environments. Moreover, structural and internal textural differences exist among various categories of aquaculture ponds, further increasing the difficulty of fine-grained classification.

Since complex boundaries and intercategory differences are primarily reflected in the frequency domain, improving edge detection and fine-grained classification relies on enhancing the model's sensitivity to frequency components. As shown in Fig. 5, wavelet decomposition effectively separates low-frequency features, which convey overall structural information, from high-frequency components that capture detailed textures and directional edges. For example, high-frequency components in horizontal, vertical, and diagonal directions clearly highlight pond boundaries and internal structures such as aerators or cages, as indicated by yellow circles. Enhancing the model's ability to perceive and utilize such frequency information is therefore essential for accurate recognition in complex aquaculture pond scenarios.

To this end, the FABC is designed and used in the BDA-Net's decoder and segmentation heads, as illustrated in Fig. 6.

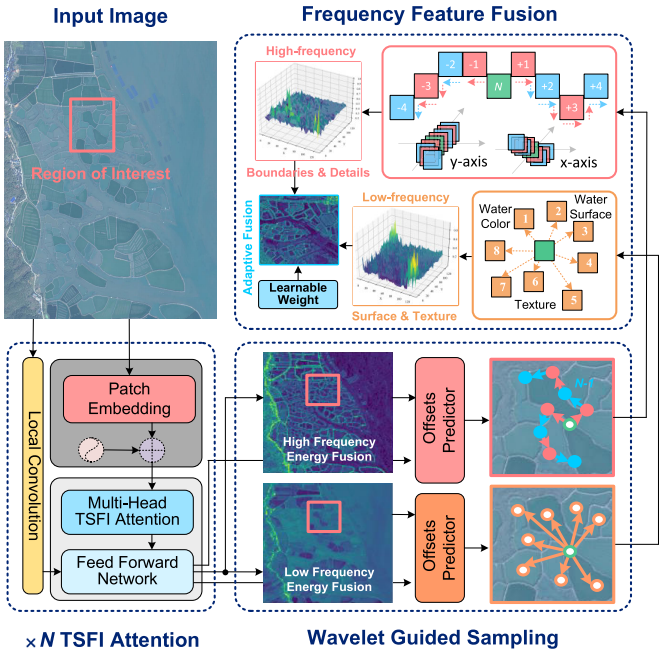


Fig. 6. Designed FABC.

In particular, it models frequency information to help the model focus on key frequency components, thereby improving its ability to recognize complex boundaries and detailed features. Equation (3) denotes the different sub-bands after wavelet decomposition. In FABC, we used the Daubechies 4 (db4) wavelet as the basis function for wavelet decomposition due to its advantages in capturing irregular boundaries and local detail features. On one hand, it has a sufficient number of vanishing moments (4th order) to effectively capture local fluctuations in boundary curves. On the other hand, its compact support property allows it to accurately represent the irregular geometric features of pond boundaries. Besides, compared to the Haar wavelet, the smoothness of db4 is more suitable for processing continuous boundaries in natural scenes. Let the input be $\boldsymbol{X}$. $\boldsymbol{X}^{(LL)}$ represents the low-frequency approximate component containing overall structural information, $\boldsymbol{X}^{(LH)}$ represents the horizontal high-frequency component emphasizing vertical edge features, $\boldsymbol{X}^{(HL)}$ the vertical high-frequency component emphasizing horizontal edge features, and $\boldsymbol{X}^{(HH)}$ represents the diagonal high-frequency component emphasizing corner and diagonal features.

First, the FABC adaptively fuses high-frequency features in different directions to determine the high-frequency direction that dominates boundary and detail learning.

$$\boldsymbol{G}^{(H)} = \text{Conv}\left(\left[\boldsymbol{X}^{(LH)}, \boldsymbol{X}^{(HL)}, \boldsymbol{X}^{(HH)}\right]\right) \tag{19}$$

$$\beta_d = \text{Softmax}\left(\boldsymbol{w}_d^{\top}\text{GAP}\left(\boldsymbol{G}^{(H)}\right) + b_d\right) \tag{20}$$

where $\beta_d, d \in \{LH, HL, HH\}$ denotes the dynamic weights of each high-frequency component. GAP$(\cdot)$ represents global average pooling. Use $\boldsymbol{G}^{(H)}$ to predict the incremental offsets of high-frequency boundaries and detailed information in different main directions of convolution sampling, that is,

$$\Delta\boldsymbol{P}^{(H)} = \sum \beta_d \cdot \text{Conv}_{\text{group}}\left(\boldsymbol{G}^{(H)}\right) = \sum \beta_d \cdot \Delta\boldsymbol{P}_d. \tag{21}$$

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Since the three high-frequency components have the same size, group convolution $\mathrm{Conv}_{\mathrm{group}}(\cdot)$ is used to improve computational efficiency. Similarly, the incremental offsets $\Delta \boldsymbol{P}^{(L)}$ predicted by the low-frequency component can also be obtained.

Different from existing DCNs, the FABC mechanism employs a recursive offset computation scheme. In this approach, the offset of each sampling position is recursively derived from the previous position's offset, then adjusted by the directional weights obtained from high- and low-frequency components. Denote the size of the convolution kernel as $K \times K$, the grid sampling points are $\{p_n\}_{n=0}^{N-1}$, where $N = K^2$, $n = 1, 2, \ldots, N-1$. For the nth point, the accumulated offset of the high-frequency sampling is

$$\Delta \boldsymbol{p}_n^{(H)} = \Delta \boldsymbol{p}_{n-1}^{(H)} + \delta \boldsymbol{p}_n^{(H)} \tag{22}$$

where $\delta \boldsymbol{p}_n^{(H)}$ is the incremental offset extracted from $\Delta \boldsymbol{P}^{(H)}$, that is, $\delta \boldsymbol{p}_n^{(H)} = \pi_n(\Delta \boldsymbol{P}(i,j)^{(H)})$, and π_n is the projection operator of $\mathbb{R}^{2N} \to \mathbb{R}^2$. The actual sampling position is

$$\boldsymbol{p}_n^{(H)} = \boldsymbol{p}_0 + \boldsymbol{p}_r + \Delta \boldsymbol{p}_n^{(H)} \tag{23}$$

where $\boldsymbol{p}_r$ represents the initial regular coordinates of the nth sampling point in the convolution kernel. For a $K \times K$ convolution kernel, $\boldsymbol{p}_r \in \{(-r,-r), (-r,-r+1), \ldots, (r,r)\}$, where $r = \lfloor K/2 \rfloor$. $\boldsymbol{p}_0$ is the spatial coordinates of the current convolution operation center (i, j).

Low-frequency features describe the overall structural information of the image. Unlike high-frequency features such as the pond's edges or internal details, they do not require as much attention to contour continuity. Therefore, the low-frequency feature sampling positions do not use the continuous recursive constraint, but instead cover a wider range of non-local areas of the image. The actual sampling position of the low-frequency feature is

$$\boldsymbol{p}_n^{(L)} = \boldsymbol{p}_0 + \boldsymbol{p}_r + \delta \boldsymbol{p}_n^{(L)} \tag{24}$$

where $\delta \boldsymbol{p}_n^{(L)} = \pi_n(\Delta \boldsymbol{P}(i,j)^{(L)})$.

Therefore, the feature of frequency-guided adaptive fusion is represented as

$$\begin{aligned} \boldsymbol{Y}\left(\boldsymbol{p}_0\right) = & \sum_{n=0}^{N-1} \alpha_n \cdot \boldsymbol{W}_n^{(L)} \cdot \boldsymbol{X}^{(LL)}\left(\boldsymbol{p}_n^{(L)}\right) \\ & + \sum_{n=0}^{N-1} (1-\alpha_n) \cdot \boldsymbol{W}_n^{(H)} \cdot \boldsymbol{G}^{(H)}\left(\boldsymbol{p}_n^{(H)}\right) \end{aligned} \tag{25}$$

where $\boldsymbol{\alpha}$ denotes the energy density balancing factor obtained by $\boldsymbol{\alpha} = \sigma(\boldsymbol{w}^\top \cdot [\boldsymbol{X}^{(LL)}, \boldsymbol{X}^{(LH)}, \boldsymbol{X}^{(HL)}, \boldsymbol{X}^{(HH)}])$, $\sigma(\cdot)$ is the sigmoid function, and $\boldsymbol{w}$ is the learnable parameter. The energy density balancing factor $\boldsymbol{\alpha}$ can adaptively adjust the contribution of high- and low-frequency features to the final result. For key fine-grained features, such as boundary contours and internal details, the weight of high-frequency feature sampling is increased. For complex background noise, the weight of high-frequency feature sampling is reduced, while low-frequency features such as smooth areas and water color gradients are emphasized.

Therefore, the designed FABC not only considers the spatial continuity characteristics of the pond boundaries but also addresses the differences in high- and low-frequency information distribution across complex environments, as well as the variations in dominant directional patterns among different pond categories.

C. Loss Functions

Due to the significant imbalance between the number of samples from aquaculture ponds and the background areas, we employ focal loss [] to supervise model training. The focal loss is represented as

$$\mathcal{L}_{\mathrm{Focal}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C} \beta_c \left(1 - q_{ic}\right)^{\gamma} y_{ic} \log\left(q_{ic}\right) \tag{26}$$

where N denotes the number of pixels and C denotes the number of categories. $q_{ic} \in (0,1)$ is the predicted probability that the pixel i belongs to class c, $y_{ic} \in \{0,1\}$ is the label assigned to class c in pixel i. β_c is the balance weight of the class ($\beta_c \propto 1/\mathrm{frequency}(c)$). γ is the focus parameter ($\gamma \geq 0$), typically $\gamma = 2$.

In addition, the model addresses the significant variation in the number of samples among different categories of ponds. For a more detailed statistical analysis of the data distribution, please refer to the experimental section. To handle these challenges, OHEM loss [] is employed for multiclass segmentation, selecting the most informative pixels.

$$\mathcal{L}_{\mathrm{OHEM}} = \frac{1}{K} \sum_{i \in \mathcal{H}} \ell_i^{\mathrm{CE}} \tag{27}$$

where $\mathcal{H}$ denotes the set of K hardest examples in the batch. $\mathcal{H} = \{i \mid \ell_i^{\mathrm{CE}} \geq \ell_{(N-K)}^{\mathrm{CE}}\}$ where $\ell_{(k)}^{\mathrm{CE}}$ is the kth order statistic. $\ell_i^{\mathrm{CE}} = -\sum_{c=1}^{C} y_{ic} \log(q_{ic})$ is the multiclass cross-entropy loss at pixel i.

Furthermore, considering that there are many small ponds and the differences between different categories of ponds are mainly concentrated in the detailed areas, Dice loss[] is introduced to improve the small object recognition performance

$$\mathcal{L}_{\mathrm{Dice}} = \frac{1}{C} \sum_{c=1}^{C} \left[1 - \frac{2 \sum_{i=1}^{N} y_{ic} q_{ic} + \epsilon}{\sum_{i=1}^{N} y_{ic}^2 + \sum_{i=1}^{N} q_{ic}^2 + \epsilon} \right] \tag{28}$$

where $\epsilon = 1 \times 10^{-5}$ is the smoothing constant.

The total loss is

$$\mathcal{L}_{\mathrm{total}} = \lambda_{\mathrm{focal}} \mathcal{L}_{\mathrm{Focal}} + \lambda_{\mathrm{ohem}} \mathcal{L}_{\mathrm{OHEM}} + \lambda_{\mathrm{dice}} \mathcal{L}_{\mathrm{Dice}} \tag{29}$$

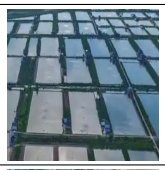
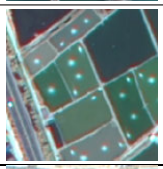
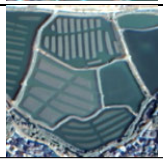
where $\lambda_{\mathrm{focal}} = 1$, $\lambda_{\mathrm{ohem}} = 0.5$, and $\lambda_{\mathrm{dice}} = 0.4$ are weighting coefficients controlling the contribution of each loss component. The focal loss down-weights well-classified pixels across all classes. The OHEM loss prioritizes challenging pixels regardless of class. The Dice loss optimizes spatial coherence for each class separately.

IV. Experiments and Discussions

In this section, we conduct extensive experiments on two datasets to comprehensively evaluate the performance of BDA-Net. We begin by introducing the experimental setup, including implementation details, datasets, and evaluation metrics. Then, ablation studies are designed to validate the

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TABLE I
DIFFERENT CATEGORIES OF AQUACULTURE PONDS

Label Type	RS Image	Field Photo	Characteristics	Proportion (%)
Normal Aquaculture Pond			The water exhibits a yellow-green or bluish-green tint, and the pond is largely free of additional equipment. Mainly used for breeding hypoxia-tolerant fish, crabs, and other species.	68.56
Pond with Aerators			High-power aerators are used, which appear as white dots in remote sensing imagery. Mainly used for breeding fish and shrimp species with high dissolved oxygen requirements.	7.66
Pond with Cages			Cage culture is used, appearing as tan-yellow striped patterns in remote sensing imagery. Mainly used for breeding sedentary shellfish such as oysters.	23.78

effectiveness of each component of BDA-Net and quantitatively assess its performance. We provide a detailed discussion on issues critical to coastal aquaculture pond extraction, such as scale adaptation, boundary extraction, and fine-grained feature capture. Furthermore, we compare the proposed model with several classical semantic segmentation methods from recent years and present a comprehensive evaluation. Finally, the generalization capability of BDA-Net is further evaluated on a larger-scale dataset.

A. Datasets

In this study, we constructed a high-resolution FGPR dataset from typical areas in Fujian Province to address two major limitations of existing datasets: low spatial resolution and the absence of subdivided pond categories. To further validate the generalization capability of the recognition model across broader geographical regions, we reprocessed the publicly available and widely used China Coastal Aquaculture Pond (CCAP) spatial distribution dataset [31], [42] (a binary-classification benchmark) to facilitate fair comparisons with other deep learning methods.

1) The FGPR dataset is constructed using GF-2 satellite imagery from 2022, with a spatial resolution of 0.8 m. It consists of 2105 image-mask pairs for training and 1120 for testing. The dataset covers three typical areas in Fujian province, China: Sansha Bay, Luoyuan Bay, and the Min River Estuary. Each image has been meticulously annotated at the pixel level and categorized into three classes: "Pond," "Pond with large aerators," and "Pond with cages." Table I provides detailed information on category definitions, image characteristics, breeding patterns, sample images, and sample distribution within the FGPR dataset. In particular, in the training set, the proportions of normal aquaculture ponds, aquaculture ponds with aerators, and aquaculture ponds with cages are 62.7%, 24.6%, and 12.7%, respectively. In the test set, the proportions for the same categories are 74.2%, 22.5%, and 3.3%. Example annotations are illustrated in Fig. 7.

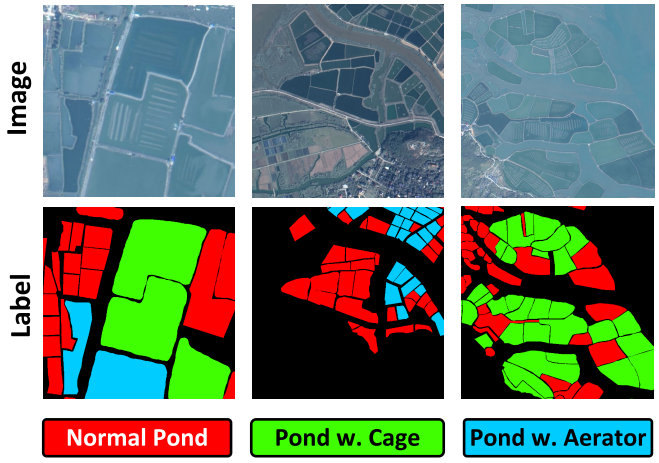


Fig. 7. Some samples and labels of the constructed FGPR dataset.

2) The CCAP dataset is a publicly available benchmark constructed using Landsat 8 imagery with a spatial resolution of 30 m [31], [42]. The original label data were provided by the Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences, and cover all coastal regions of China. Although originally designed for traditional mapping purposes rather than deep learning-based evaluation, we reprocessed this dataset through relabeling and structural optimization to establish a fair and effective benchmark for comparative analysis of deep learning methods. Due to the relatively low resolution of the source images, the labels include only two categories: "Pond" and "Background." The reprocessed training and test sets

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Fig. 8. Environmental differences in aquaculture ponds in different regions of China.

TABLE II
SAMPLE DISTRIBUTION OF EACH AREA IN THE CHINA AQUACULTURE POND DATASET

Regions	Sample Proportion (%)
Fujian & Taiwan	5.919
Guangdong	9.279
Guangxi	6.162
Hebei & Tianjin	18.997
Hainan	10.109
Jiangsu	10.596
Liaoning	11.085
Shandong	14.047
Shanghai	6.466
Zhejiang	4.127
Others	3.213
Total	100

contain 9537 and 4088 image-mask pairs, respectively. To enhance the assessment of model generalization, the training and test samples are explicitly divided across different geographic regions with significant variations in environmental conditions and water color. We provide a statistical analysis of the distribution of positive and negative samples for each province in Table II, and further illustrate the diversity in water color across different marine regions with sample images in Fig. 8.

To increase the size of the datasets, both were augmented in advance using techniques such as flipping, adding noise, and adjusting contrast and saturation. During the splitting of the training and validation sets, we ensured that the proportion of each category closely matched the overall distribution.

B. Experimental Conditions

1) Experimental Setting: The experiments are performed on a server with four NVIDIA GeForce RTX 4090 GPUs (24 GB). The AdamW optimizer was adopted for its adaptability, with learning rate set to 6×10^{-4} and weight decay set to 0.01 via grid search. The maximum training epoch is 150 to accommodate the full convergence potential of models. To ensure fairness, identical hyperparameters were applied to all baseline methods (e.g., UNetFormer), eliminating tuning biases.

2) Evaluation Metrics: This study employs four widely used metrics for fine-grained semantic segmentation evaluation: mean intersection over union (mIoU), frequency weighted intersection over union (FWIoU), overall accuracy (OA), and the $F1$-measure, in addition to the inference speed metric frames per second (FPS).

mIoU is a standard benchmark in deep learning that computes the average ratio of the intersection to the union across all categories. FWIoU extends mIoU by incorporating class frequency weights, thereby offering better performance assessment in fine-grained scenarios where class distributions are imbalanced. OA measures the proportion of correctly classified pixels out of the total number of pixels. The $F1$-measure represents the harmonic mean of precision and recall, with higher values indicating greater model robustness. Both OA and $F1$-measure are commonly adopted in semantic segmentation tasks.

C. Ablation Experiments

This study adopts UNetFormer as the baseline model, which utilizes ResNet18 as the backbone and integrates a Transformer module into the encoder. To systematically evaluate the contribution of each component, a series of ablation experiments was conducted. The results of these experiments for different modules of BDA-Net are summarized in Table III. The proposed TSFI and FABC modules improved the model's mIoU on the dataset by 3.54% and 1.87%, respectively. The complete BDA-Net model achieved gains of 5.45% in mIoU, 3.49% in $F1$-score, 1.40% in OA, and 2.80% in FWIoU compared to the baseline UNetFormer [20], demonstrating consistently strong performance across all metrics.

Furthermore, we compared TSFI with the existing multiscale attention method PVT [43], and FABC with other local detail convolution methods, including DCN [44] and DSC [23]. As shown in Table III, TSFI achieved superior performance over PVT [43], with mIoU increased by 0.29% (82.13% versus 81.84%). Similarly, FABC outperformed both DCN and DSC, improving mIoU by 1.15% and 1.04% over DSC [23] and DCN [44], respectively. These results validate the advantages of the proposed modules over existing alternatives in handling scale variation and boundary details.

Besides, ablation studies demonstrate a well-balanced performance-computation tradeoff achieved by the TSFI and FABC modules. The TSFI module demonstrates remarkable efficiency, achieving 52.48 FPS with a + 1.87% mIoU gain through its Taylor approximation design. Subsequent

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TABLE III
ABLATION STUDY OF EACH COMPONENT OF THE BDA-NET

Methods	Win. Att.	PVT[43]	TSFI	DCN[44]	DSC[23]	FABC	mIoU(%)	F1(%)	OA(%)	FWIoU(%)	Speed (fps)
UNetFormer[20] (Baseline)	✓						80.26	88.64	94.58	90.01	71.19
+ Global Multi-scale Attention		✓					81.84	89.77	94.60	90.08	21.07
			✓				82.13	89.93	94.62	90.16	52.48
+ Local Detail Convolution	✓			✓			81.76	89.98	95.03	91.02	65.51
	✓				✓		82.65	90.19	95.21	91.00	67.29
	✓					✓	83.80	90.95	95.37	91.17	63.34
Ours			✓			✓	**85.71**	**92.13**	**95.98**	**92.81**	49.26

Note: Bold indicates the best results. "Win. Att." denotes the window attention used in UNetFormer[20]. "PVT" denotes using the multi-scale pyramid vision Transformer[43]. "DCN" and "DSC" denote using the deformable convolution[44] and dynamic snake convolution[23], respectively.

TABLE IV
ABLATION STUDY OF THE TAYLOR-OPTIMIZED ALGORITHM

Methods	G/L	Multi-Scale	mIoU(%)	F1(%)	OA(%)	FWIoU(%)	Computational Complexity	Memory Complexity	FPS
UNetFormer[20]	Local	×	80.26	88.64	94.58	90.01	$O((n/w)^2 d_k n_w)$	$O((n/w)^2 n_w)$	71.19
UNetFormer[20]	Global	×	81.33	89.04	94.70	90.12	$O(n^2 d_k)$	$O(n^2)$	41.65
w/o. Taylor-optimized	Global	✓	83.18	90.60	94.91	90.39	$O(nC + n^2 d_k / L)$	$O(n^2)$	38.12
w. Taylor-optimized	Global	✓	82.13	89.93	94.62	90.16	$O(nC + L\dot{(}nd_k^2 + k\dot{n}d_k r))$	$O(n(d_k + r))$	52.48

Note: "Multi-Scale" denotes whether using multi-scale self-attention. "G/L" denotes using global attention or local windowed attention. $L = 4$ is the number of wavelet scales, r is the projection dimension ($r \ll d_k \ll n$), C denotes the number of channels. w is the window size, n_w is the number of local windows.

integration with FABC delivers an additional + 3.58% mIoU improvement over TSFI alone while maintaining 49.26 FPS, yielding an optimal balance of 85.71% mIoU accuracy with efficient computational costs. This synergy confirms that both modules provide exceptional value relative to resource requirements.

As illustrated in the visualization results provided in Fig. 9, the proposed modules contribute significantly to improving pond segmentation quality. The TSFI module effectively addresses scale mismatches by capturing multiscale contextual information, leading to more consistent and complete pond segmentation compared to the baseline, especially in regions with large size variations. The FABC module enhances boundary integrity and fine-grained detail preservation through adaptive fusion of frequency-aware features, resulting in sharper contours and reduced ambiguity along edges. As seen from the feature visualization in Fig. 10, the TSFI module's multiscale contextual features provide robust region proposals by capturing pond size variations, which the FABC module then refines through boundary-aware frequency decomposition to sharpen contours and resolve ambiguities. Overall, the BDA-Net integrates both components synergistically, achieving the best visual accuracy with clearer structures and more coherent regional segmentation. Next, we further analyze the proposed TSFI and FABC.

1) Analysis of the TSFI Structure: The TSFI module adaptively downsamples the $\boldsymbol{Q}$ and $\boldsymbol{K}$ tensors by multilevel wavelet decomposition, enabling the model to capture receptive fields of varying sizes. This design overcomes the constraint of fixed-scale receptive fields in standard attention mechanisms, significantly enhances the model's ability to represent objects across different scales, and increases its adaptability to aquaculture ponds with large morphological variations. To further validate the effectiveness of the TSFI module, we compare the feature maps with and without its incorporation, as visualized in Fig. 11. The results clearly demonstrate that the TSFI module substantially enhances the model's capacity for scale-aware feature extraction. It not only improves detection sensitivity for smaller ponds but also refines the recognition accuracy of larger ones, effectively reducing recognition errors across diverse pond categories.

Furthermore, to evaluate the effectiveness of the Taylor-optimized strategy in reducing computational complexity of multiscale self-attention, we conducted a dedicated ablation study using FPS as the primary speed metric. As shown in Table IV, introducing the multiscale attention mechanism without Taylor optimization ("without Taylor-optimized") significantly reduces inference speed, with FPS dropping from 71.19 to 38.12 compared to the UNetFormer baseline. This confirms the substantial

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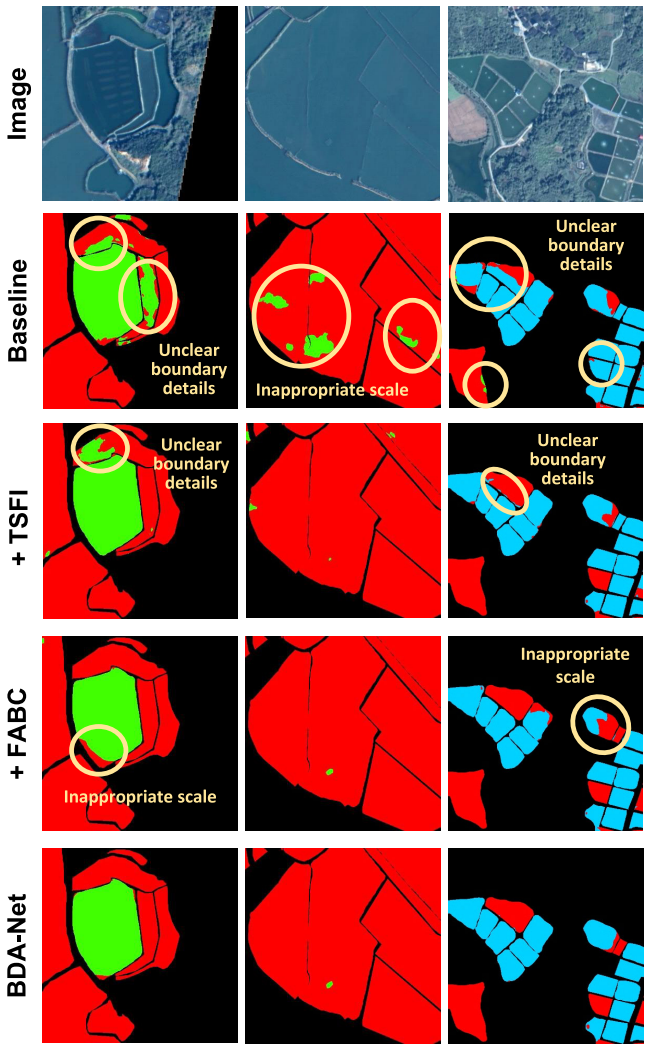


Fig. 9. Visualization results of the ablation study. The yellow circles mark some key feature regions for comparison.

computational overhead of standard multiscale self-attention design.

After incorporating Taylor-based dimensionality reduction ("with Taylor-optimized"), the model achieves a notable recovery in inference speed, reaching 52.48 FPS. This represents an improvement of 14.36 over the nonoptimized version while maintaining competitive segmentation accuracy. The mIoU decreased by only about 1%, with other metrics fluctuating within 1%, demonstrating that the Taylor approximation effectively reduces computational overhead with negligible performance impact. All experiments were conducted under identical software and hardware conditions, ensuring fair comparison.

2) Analysis of the FABC Module: We employ the FABC as a feature enhancement filter to improve the model's ability to recognize complex and irregular boundaries. The FABC operates along both the x- and y-axes, extracting directional boundary features that correspond to horizontal and vertical structures, respectively. During the experiments, we visualized the outputs of FABC in both directions, as illustrated in

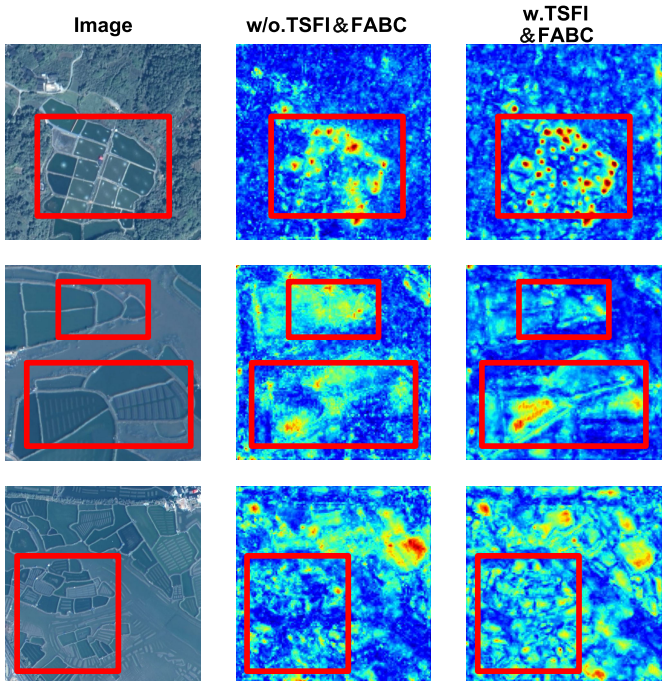


Fig. 10. Feature visualization results of the combined effects of TSFI and FABC. The red boxes mark some key feature regions for comparison.

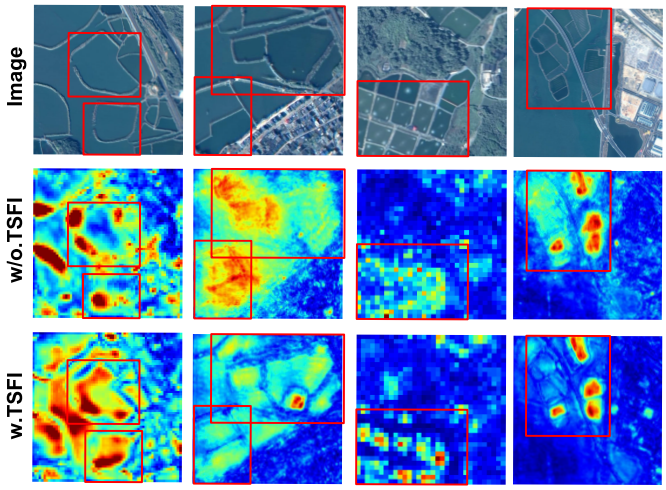


Fig. 11. Visualization of the TSFI feature maps. The red boxes mark some key feature regions for comparison.

Fig. 12. The results show that the extracted features align accurately with the actual orientation of the aquaculture ponds' boundaries, confirming FABC's capability to capture structural characteristics and adaptively guide the convolutional kernel in handling intricate contour information.

As shown in Fig. 5, images inherently contain rich frequency-domain information. However, as network depth increases, most deep learning models suffer from a gradual loss of high-frequency details (see Fig. 13), which limits their ability to represent fine-grained features. To address this issue, we integrate a wavelet decomposition algorithm into the FABC module. This algorithm explicitly decomposes and utilizes both high- and low-frequency features, allowing the model to better preserve and enhance structural details, e.g., edges and texture variations, in aquaculture ponds.

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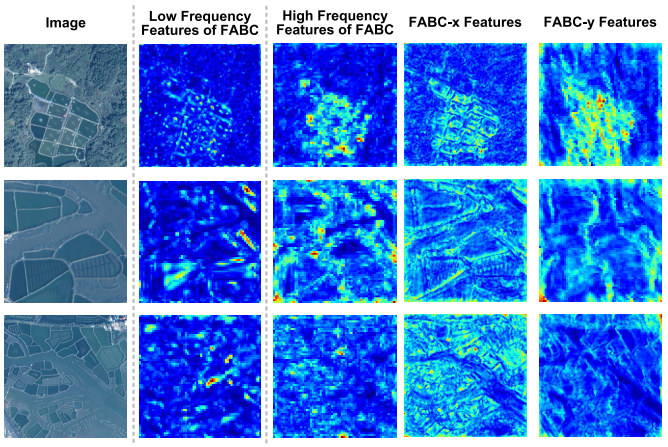


Fig. 12. Visualization of the FABC's features: low-frequency, high-frequency, and those oriented along the X- and Y-axes.

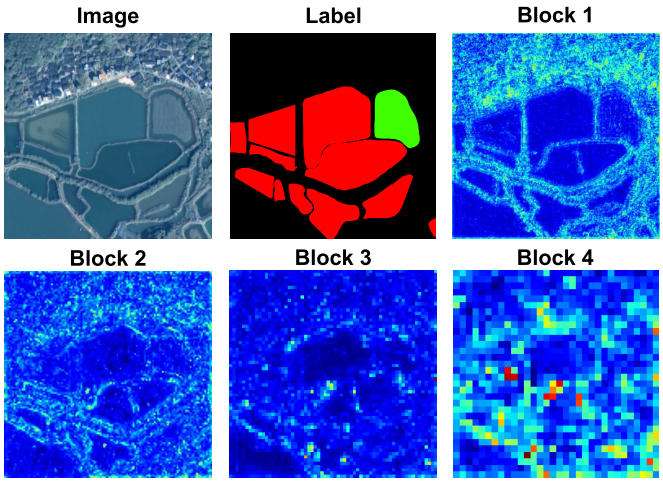


Fig. 13. Frequency domain features from different layers of the model. As the number of model layers increases, high-frequency detail information is gradually lost.

To further evaluate the contribution of frequency guidance, we conducted an ablation study comparing the baseline model, the FABC module without wavelet decomposition, and the full FABC module with wavelet guidance. As shown in Fig. 14, the baseline model (second column) exhibits significant loss of boundary details, resulting in blurred and incomplete feature representations. Introducing FABC without wavelet decomposition (third column) partially enhances the model's ability to focus on relevant regions, yet the extracted features remain unclear and susceptible to interference from surrounding structures. In contrast, the complete FABC module incorporating wavelet-based frequency guidance (fourth column) produces sharply defined and highly discriminative feature maps, accurately capturing fine details such as aquaculture cages and aerators. These results underscore the critical role of wavelet-guided frequency decomposition in enabling robust feature extraction and improving fine-grained classification performance for complex aquaculture pond boundaries.

In addition, Table V compares the performance of FABC using different wavelet basis functions. The experimental results show that using the Daubechies 4 (db4) wavelet achieved the best performance across all metrics,

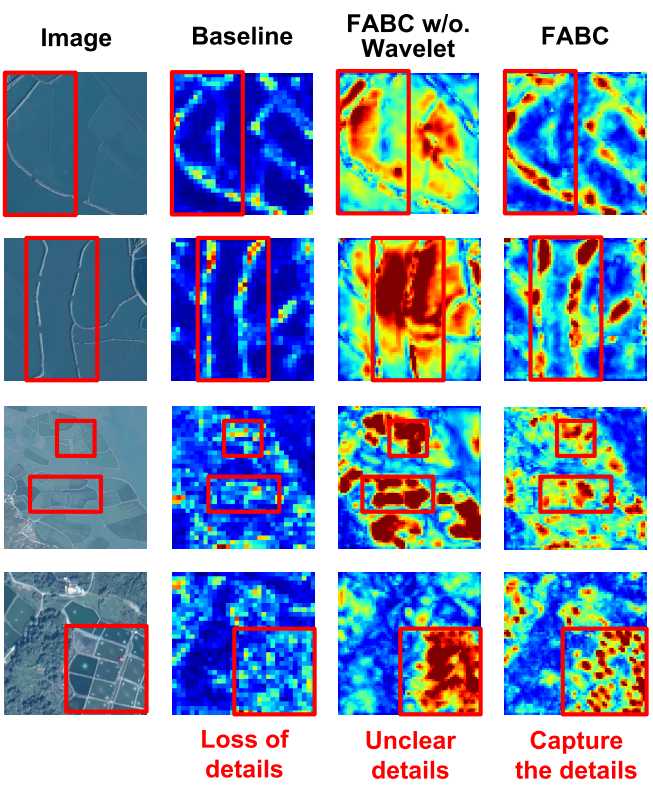


Fig. 14. Visualization of the FABC feature maps. The red boxes mark some key feature regions for comparison.

TABLE V
ABLATION STUDY OF THE FABC WITH DIFFERENT WAVELETS

Module	Wavelets	mIoU(%)	F1(%)	OA(%)	FWIoU(%)
Baseline	w./o.	81.76	89.98	95.03	91.02
+FABC	Haar	82.88	90.34	95.23	91.10
+FABC	Symlet	83.51	90.87	95.30	91.09
+FABC	Daubechies 4	83.80	90.95	95.37	91.17

outperforming the Haar wavelet and the Symlet wavelet. This performance advantage is attributed to the optimal balance of four vanishing moments and compact support in db4, enabling it to more effectively capture the complex and irregular boundary features unique to aquaculture ponds.

D. Comparison Experiments

To evaluate the effectiveness of the proposed BDA-Net, we compared it with the state-of-the-art (SOTA) semantic segmentation methods: A^{2}FPN [45], BANet [46], DC-Swin [47], ABCNet [48], MANet [49], and the classic U-Net [50].

A^{2}FPN [45] introduces attention enhancement and semantic enhancement modules to achieve adaptive multiscale feature fusion and improve semantic consistency, thereby enhancing segmentation accuracy for small objects and boundaries in remote sensing imagery. BANet [46] combines Transformer and convolutional branches with a bidirectional awareness mechanism, enabling effective modeling of both local and global semantic information in high-resolution urban scenes.

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TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT SEMANTIC SEGMENTATION METHODS ON FUJIAN DATASET

Methods	Backbone	Normal Pond(%)	w. Aerators(%)	w. Cages(%)	mIoU(%)	F1(%)	OA(%)	FWIoU(%)	Parameters(M)	GFLOPs	Speed(fps)
A²FPN[45]	ResNet18	93.41	69.93	83r.22	82.19	89.91	94.85	90.35	12.2	672.6	35.42
ABCNet[48]	ResNet18	92.29	60.13	79.06	77.16	86.46	93.80	88.52	13.9	370.7	42.77
BANet[46]	ResNet18	93.94	67.20	83.50	81.55	89.42	95.17	90.92	12.7	77.5	52.21
DC-Swin[47]	Swin-T	92.87	72.62	78.96	81.48	89.56	94.28	89.30	118.4	760.7	15.56
MANet[49]	ResNet50	90.47	60.75	73.96	75.06	85.20	92.31	85.78	35.9	1088.3	23.90
UNet[50]	VGG16	91.44	41.73	80.58	71.25	81.22	93.18	87.37	12.7	77.5	76.64
UNetFormer[20]	ResNet18	92.75	66.38	81.64	80.26	88.64	94.58	90.01	11.9	360.6	71.19
Ours	ResNet18	**94.79**	**75.81**	**86.55**	**85.71**	**92.13**	**95.98**	**92.81**	12.2	436.9	49.26

Note: Bold indicates the best results.

DC-Swin [47] employs a Transformer-based architecture augmented with local detail modeling and global contextual perception, supporting fine-grained semantic segmentation of complex land cover types in high-resolution remote sensing images. ABCNet [48] utilizes an attention-guided bidirectional context interaction module to strengthen the coupling of spatial and semantic information, improving both efficiency and accuracy in segmenting target regions. MANet [49] constructs a multilevel attention fusion network to selectively aggregate multiscale and multisemantic features, significantly improving segmentation performance for complex ground objects.

The comparative results are summarized in Table VI. Overall, the proposed BDA-Net outperforms all baseline methods across all evaluation metrics. Compared to the UNetFormer baseline, it achieves improvements of 5.45% in mIoU, 3.49% in $F1$-score, 1.40% in OA, and 2.80% in FWIoU. In particular, BDA-Net attains an mIoU of 85.71%, exceeding A^2FPN, ABCNet, BANet, DC-Swin, and MANet by 3.52%, 8.55%, 4.16%, 4.23%, and 10.65%, respectively. All of these comparison models are based on attention mechanisms. This result demonstrates BDA-Net's higher segmentation accuracy and improved recognition performance for aquaculture ponds. Furthermore, BDA-Net achieves an FWIoU of 92.81%, reflecting its strong capability in fine-grained classification, particularly under class-imbalanced conditions. Furthermore, as evidenced in Table VI, our method achieves the best performance balance by attaining the highest mIoU (85.71%) while maintaining competitive efficiency. Compared to other methods, our approach yields a 2.85% mIoU improvement over the second-best method (BANet: 81.55%) with only a moderate speed reduction. This demonstrates that the TSFI and FABC modules provide substantial accuracy gains while keeping computational costs within practical limits for large-scale deployment. The 49.26 frames/s inference speed further confirms the model's suitability for real-time applications.

Furthermore, we present visual comparisons of the experimental results on the FGPR dataset in Fig. 15. These comparisons further validate the effectiveness of the proposed BDA-Net. In the first sample group, only A^2FPN [45], U-Net [50], BANet [46], and our BDA-Net correctly segment all three pond categories. Other models misclassify certain areas, especially confusing "Pond with Cage" (green) with "Normal Pond" (red). In the second sample group, although DC-Swin [47] captures the overall shape of the ponds, its boundaries are blurry, and it fails to distinguish "Pond with Cage" from "Normal Pond" inside the yellow rectangular region. In contrast, BDA-Net not only accurately outlines the ponds but also correctly identifies the cage structures inside. Similar improvements are observed in the third and fourth sample groups. BDA-Net maintains clearer boundaries and more consistent categorization in dense pond areas, whereas other models exhibit noticeable fragmentation or merging errors. For example, in the yellow rectangular region of the fourth sample, MANet merges adjacent ponds of different categories, while BDA-Net preserves clear separations.

In summary, while existing deep learning methods can effectively recognize individual categories of aquaculture ponds, their performance declines as the number of categories increases. Most models struggle to accurately capture distinctive features and fine-grained differences among pond categories, leading to confusion in both categorical prediction and boundary localization. This limitation is most pronounced in DC-Swin [47], which performs well in simple scenes but shows significant errors when distinguishing multiple pond categories in complex environments. Other models are also affected to varying degrees. In contrast, the proposed BDA-Net delivers more consistent and robust segmentation results. By incorporating the TSFI and FABC modules, our approach effectively models scale variation, boundary details, and structural characteristics of aquaculture ponds. The method substantially improves segmentation accuracy and stability, particularly in challenging nearshore environments with complex backgrounds and diverse pond categories.

In addition, to evaluate the model's robustness to resolution variations, we down-sampled the high-resolution FGPR images (0.8 m) to multiple intermediate resolutions (1.6, 3.2, 6.4, and 12.8 m) and evaluated the performance trend across this controlled gradient. The results shown in Table VII reveal two key observations: First, fine-grained multiclass segmentation ("-mc") exhibits significant performance degradation as resolution decreases (e.g., BDA-Net drops from

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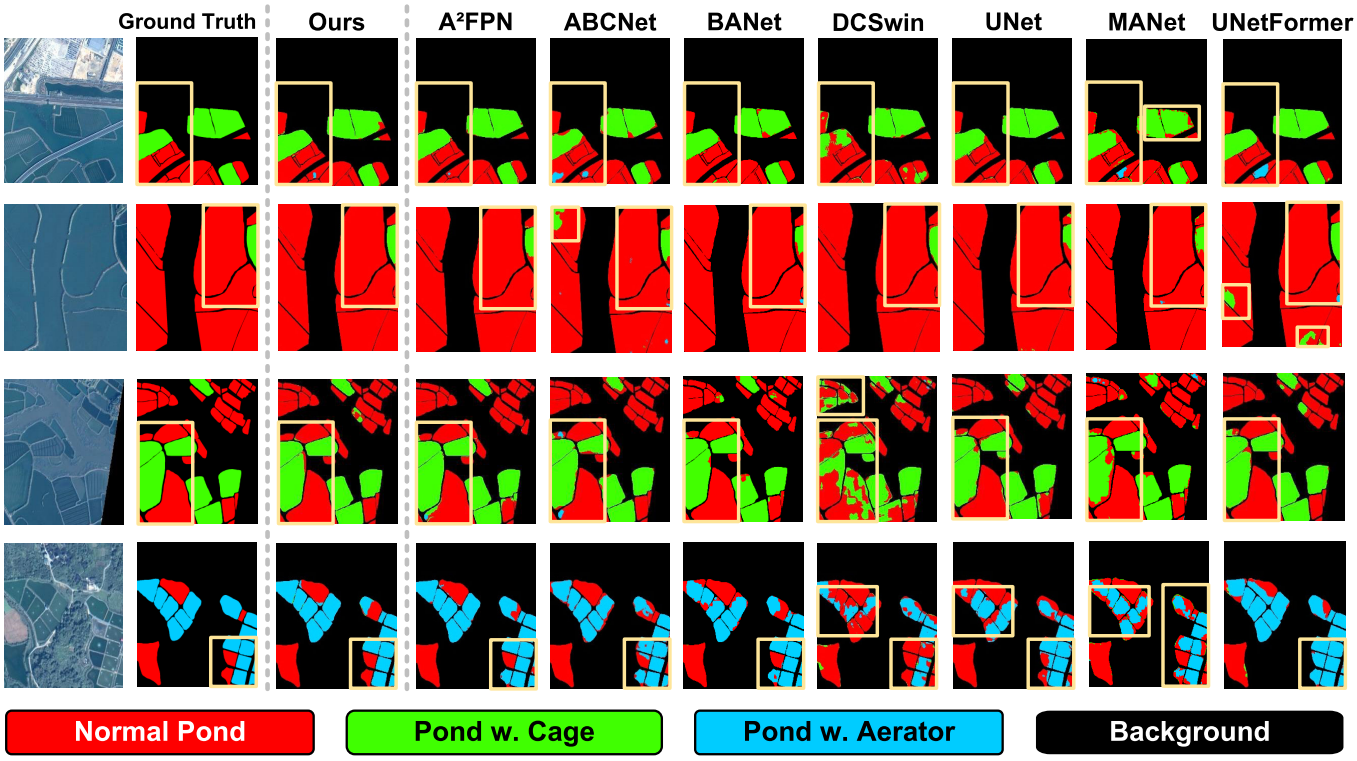


Fig. 15. Comparison experiments on the constructed FGPR dataset. The yellow boxes mark some key feature regions for comparison.

TABLE VII
EXPERIMENTS ON FGPR IMAGES AT DIFFERENT RESOLUTIONS

Spatial resolution	Baseline (0.8m)	×1/2	×1/4	×1/8	×1/16
BDA-Net mIoU(%)-mc	85.71	77.85	71.91	52.31	35.47
UNetFormer mIoU(%)-mc	80.26	74.89	67.97	51.77	28.31
BDA-Net mIoU(%)-bc	90.12	88.74	82.39	78.86	75.07
UnetFormer-Net mIoU(%)-bc	87.65	84.10	77.45	71.28	69.03

Note: "-mc" indicates multi-class classification, and "-bc" indicates binary classification.

85.71% to 35.47% at 12.8 m), whereas binary classification ("-bc") remains relatively stable (90.12% to 75.07%) since it only distinguishes ponds from background. Second, BDA-Net demonstrates superior robustness compared to UNetFormer, particularly in challenging scenarios—at 12.8-m resolution, BDA-Net achieves 7.16% higher mIoU than UNetFormer in multiclass segmentation, confirming its better adaptability to resolution variations. These findings solidly support our claims about the model's applicability across diverse resolution conditions.

E. Generalization Performance

The irregular boundaries of aquaculture ponds and the variability of aquatic environments present considerable challenges to model generalization. To assess how well different methods adapt to aquaculture ponds in diverse regions and water conditions, we compared the segmentation performance of all models on a nationwide dataset. This experiment uses the CCAP dataset as an independent test benchmark, which contains samples of nearshore aquaculture ponds from coastal areas across China. The quantitative results are provided in Table VIII.

On this dataset, the proposed BDA-Net achieved a pond recognition accuracy of 73.00% and a background classification accuracy of 96.92%. It also attained high scores across other key metrics, including 84.81% mIoU, 91.33% $F1$-score, 96.90% OA, and 94.23% FWIoU, reflecting its superior overall segmentation performance. In comparison, other models yielded consistently lower results. For instance, UNetFormer [20] attained only 70.45% in pond recognition accuracy, while ABCNet [48] reached 70.28% for ponds and 96.24% for the background. Our method outperforms not only the lightweight ABCNet [48] but also the Transformer-based DC-Swin [47], surpassing the latter by margins of 3.57% in pond recognition accuracy, 0.43% in background accuracy, 2.03% in mIoU, and 1.32% in $F1$-score.

The results demonstrate that the proposed model achieves excellent performance in pond segmentation, surpassing other methods even in completely unfamiliar geographical regions. These findings not only confirm its effectiveness across diverse geographic conditions but also underscore its potential for identifying ponds in unseen environments, further validating the model's strong generalization and transfer capabilities in semantic segmentation. In addition, we provide visualizations of the experimental results, as illustrated in Fig. 16. The proposed method exhibits improved adaptability to fragmented pond boundaries and achieves higher accuracy in detecting small-sized aquaculture ponds, with fewer false positives and missed detections.

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TABLE VIII
PERFORMANCE COMPARISON OF DIFFERENT SEMANTIC SEGMENTATION METHODS ON THE 2015 CHINA POND DATASET

Methods	Backbone	Pond(%)	Background(%)	mIoU(%)	F1(%)	OA(%)	FWIoU(%)	Parameters(M)	GFLOPs	Speed(fps)
A²FPN[45]	ResNet18	70.45	96.15	83.34	90.35	96.47	93.55	12.2	672.6	36.00
ABCNet[48]	ResNet18	70.28	96.24	83.26	90.31	96.55	93.62	13.9	370.7	43.54
BANet[46]	ResNet18	68.43	96.01	82.22	89.61	96.33	93.22	12.7	77.5	52.46
DC-Swin[47]	Swin-T	69.43	96.19	82.81	90.01	96.49	93.49	118.4	760.7	15.89
MANet[49]	ResNet50	52.55	93.74	73.15	82.83	94.15	89.58	35.9	1088.3	24.07
UNet[50]	VGG16	56.30	94.38	75.34	84.57	94.76	90.53	12.7	77.5	77.19
UNetFormer[20]	ResNet18	70.45	96.15	83.30	90.35	96.47	93.55	11.9	360.6	71.22
Ours	ResNet18	**73.00**	**96.62**	**84.81**	**91.33**	**96.90**	**94.23**	12.2	436.9	50.77

Note: Bold indicates the best results.

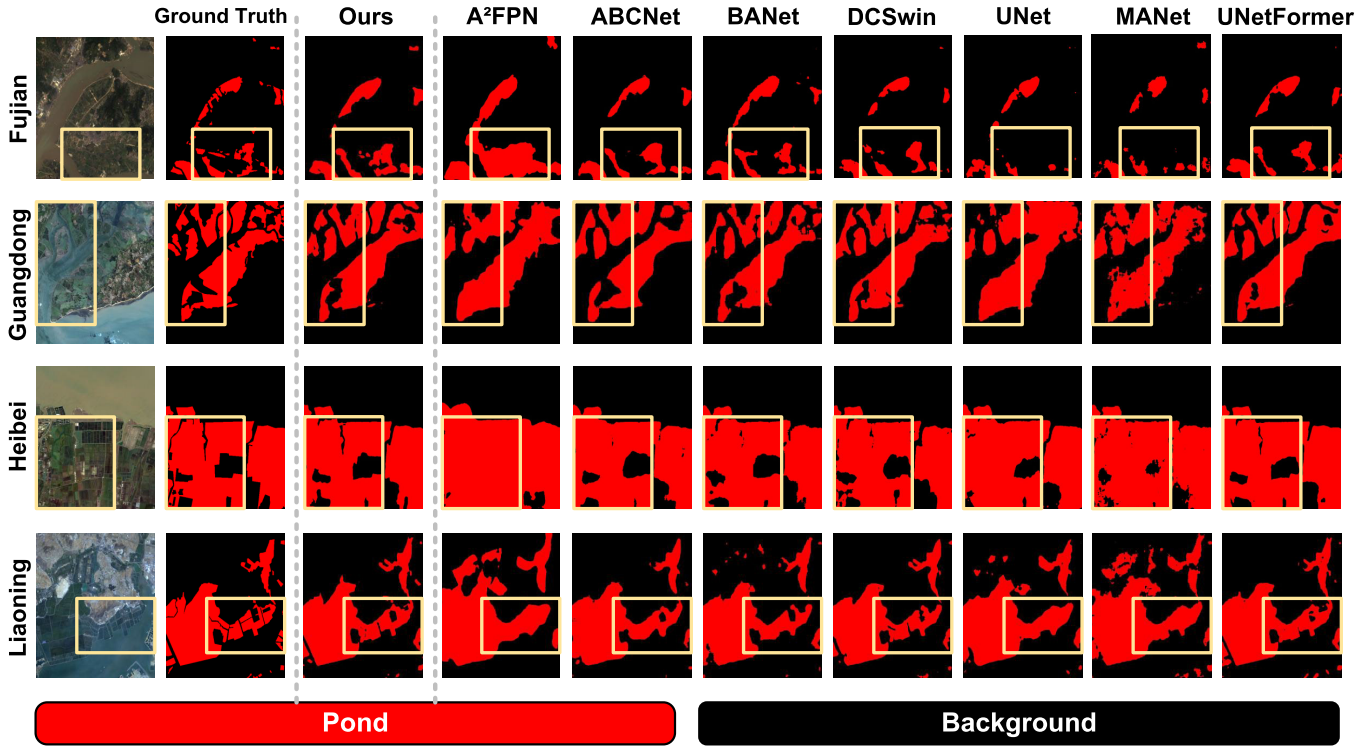


Fig. 16. Comparison experiments on China datasets. The yellow boxes mark some key feature regions for comparison.

V. CONCLUSION

This article presents a high-resolution FGPR dataset, constructed to address key limitations in existing datasets, such as low spatial resolution and the absence of detailed category subdivisions for coastal aquaculture ponds. To tackle the corresponding extraction challenges, we propose BDA-Net, a novel network that incorporates a TSFI structure to capture multiscale pond features, along with an FABC module to enhance the recognition of irregular boundaries and fine details. Extensive experiments on two datasets demonstrate that as follows.

1) Each component of BDA-Net contributes effectively to the model's performance, with all modules thoroughly validated.
2) BDA-Net outperforms current SOTA semantic segmentation models, achieving an mIoU of 85.71%, which is 5.45% higher than existing mainstream methods.
3) On a larger CCAP dataset with wider coverage, BDA-Net achieves SOTA performance, confirming its strong generalization capability and broad applicability.

Currently, due to data limitations, the proposed method cannot yet cover all types of aquaculture ponds, and the model complexity still needs further optimization to enable lightweight deployment. Future work will focus on improving the efficiency of BDA-Net and extending its application to identify a wider variety of aquatic species.

The code and datasets will be publicly available at https://github.com/Shank2358/BDANet

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APPENDIX

The errors in the proposed TSFI arise from two components: the Taylor truncation error incurred when approximating the exponential function, and the low-rank projection error introduced by the projection matrix used for computational acceleration. The bounds for these errors are derived below.

A. Taylor Truncation Error

Theorem 1 (Scalar Taylor Remainder): For any real number $x \in \mathbb{R}$ and positive integer $k \in \mathbb{Z}^+$, according to Taylor's remainder theorem (Lagrange form), the Taylor approximation error for the exponential function satisfies

$$\left|\exp(x) - \sum_{m=0}^{k} \frac{x^m}{m!}\right| \le \frac{e^{|x|}|x|^{k+1}}{(k+1)!}. \tag{A-1}$$

1) Rowwise Taylor Truncation Error: Define $\boldsymbol{A}^{(l)} = \text{softmax}(\boldsymbol{T}^{(l)})$ as the attention matrix with $\boldsymbol{T}^{(l)} = \boldsymbol{Q}(\boldsymbol{K}^{(l)})^\top/(d_k)^{1/2}$. For any row vector $\boldsymbol{t}_i \in \mathbb{R}^{1\times N}$ of $\boldsymbol{T}^{(l)}$ and the corresponding value matrix $\boldsymbol{V}^{(l)}$, apply Theorem 1 to each element t_{ij}

$$\left|\exp(t_{ij}) - \sum_{m=0}^{k} \frac{t_{ij}^m}{m!}\right| \le \frac{e^{|t_{ij}|}\,|t_{ij}|^{k+1}}{(k+1)!} \le \frac{e^{\|\boldsymbol{t}_i\|_\infty}\,\|\boldsymbol{t}_i\|_\infty^{k+1}}{(k+1)!}. \tag{A-2}$$

Let $\boldsymbol{a}_i = \text{softmax}(\boldsymbol{t}_i)$ and $\tilde{\boldsymbol{a}}_i = \text{normalize}\left(\sum_{m=0}^{k}(\boldsymbol{t}_i^{\otimes m})/(m!)\right)$, where normalize(.) denotes division by the sum to ensure a probability distribution. Since normalization is a division operation, its error amplification is an upper bound. The normalization operation in softmax is Lipschitz continuous (with the Lipschitz constant dependent on the input range). Therefore, the overall error is controlled by a linear combination of element errors. Combining with the triangle inequality, the row output error satisfies

$$\begin{aligned}\left|\boldsymbol{a}_i\boldsymbol{V}^{(l)} - \tilde{\boldsymbol{a}}_i\boldsymbol{V}^{(l)}\right| &\le \max_j \left|\exp(t_{ij}) - \sum_{m=0}^{k}\frac{t_{ij}^m}{m!}\right| \cdot \sum_j |v_{ij}| \\ &\le \frac{e^{\|\boldsymbol{t}_i\|_\infty}\,\|\boldsymbol{t}_i\|_\infty^{k+1}}{(k+1)!}\left\|\boldsymbol{V}^{(l)}\right\|_F.\end{aligned} \tag{A-3}$$

2) Matrix-Level Taylor Truncation Error: The Frobenius norm is defined as the Euclidean sum of row norms, i.e., $\|\boldsymbol{X}\|_F^2 = \sum_i \|\boldsymbol{x}_i\|_2^2$, where $\boldsymbol{x}_i$ is the ith row of $\boldsymbol{X}$. Applying A-3 rowwise and summing over all rows

$$\begin{aligned}\mathcal{E}_{\text{Taylor}}^2 &= \sum_{i=1}^{N}\left\|\boldsymbol{a}_i\boldsymbol{V}^{(l)} - \tilde{\boldsymbol{a}}_i\boldsymbol{V}^{(l)}\right\|_2^2 \\ &\le \sum_{i=1}^{N}\left(\frac{e^{\|\boldsymbol{t}_i\|_\infty}\,\|\boldsymbol{t}_i\|_\infty^{k+1}}{(k+1)!}\right)^2\left\|\boldsymbol{V}^{(l)}\right\|_F^2\end{aligned} \tag{A-4}$$

where $\mathcal{E}_{\text{Taylor}}$ denotes the Taylor truncation error.

Since $\|\boldsymbol{t}_i\|_\infty \le \|\boldsymbol{T}^{(l)}\|_F$ (as the Frobenius norm dominates the infinity norm) and $e^{\|\boldsymbol{t}_i\|_\infty} \le M$, we have

$$\begin{aligned}\mathcal{E}_{\text{Taylor}} &\le \frac{M}{(k+1)!}\left(\sum_{i=1}^{N}\|\boldsymbol{t}_i\|_\infty^{2(k+1)}\right)^{1/2}\left\|\boldsymbol{V}^{(l)}\right\|_F \\ &\le \frac{M}{(k+1)!}\left\|\boldsymbol{T}^{(l)}\right\|_F^{k+1}\left\|\boldsymbol{V}^{(l)}\right\|_F\end{aligned} \tag{A-5}$$

where $\sum_i \|\boldsymbol{t}_i\|_\infty^{2(k+1)} \le (\max_i \|\boldsymbol{t}_i\|_\infty^{k+1})^2 N \le \|\boldsymbol{T}^{(l)}\|_F^{2(k+1)}$. For any target error $\epsilon_1 > 0$, to ensure $\mathcal{E}_{\text{Taylor}} \le \epsilon_1\|\boldsymbol{V}^{(l)}\|_F$, the Taylor order k must satisfy

$$\frac{M}{(k+1)!}\left\|\boldsymbol{T}^{(l)}\right\|_F^{k+1} \le \epsilon_1. \tag{A-6}$$

B. Low-Rank Projection Error

The low-rank projection error arises from using a projection matrix $\boldsymbol{P} \in \mathbb{R}^{d_k\times r}$ (where the rank $r \ll d_k$) to accelerate the computation of high-order terms, which introduces Johnson–Lindenstrauss approximation error [41].

Lemma 1 (Johnson–Lindenstrauss Lemma): For any matrix $\boldsymbol{M} \in \mathbb{R}^{d\times n}$ and error parameter $\delta > 0$, there exists a random projection matrix $\boldsymbol{P} \in \mathbb{R}^{d\times r}$ with projection dimension $r \ge C\log(1/\delta)$ (where C is a constant), such that with high probability

$$\left\|\boldsymbol{M}^\top\boldsymbol{P}\boldsymbol{P}^\top\boldsymbol{M} - \boldsymbol{M}^\top\boldsymbol{M}\right\|_F \le \delta\|\boldsymbol{M}\|_F^2 \tag{A-7}$$

where $\|\cdot\|_F$ denotes the Frobenius norm.

1) Low-Rank Projection Error for Second-Order Term: The exact second-order term [see 10] is given by

$$\boldsymbol{A}_{(2)}^{(l)}\boldsymbol{V}^{(l)} = \frac{1}{2}\left(\boldsymbol{Q}\boldsymbol{K}^{(l)\top}\right)^{\odot 2}\boldsymbol{V}^{(l)} \tag{A-8}$$

where $\odot 2$ denotes the Hadamard power [40]. Using tensor product expansion, we have

$$\left(\boldsymbol{Q}\boldsymbol{K}^{(l)\top}\right)^{\odot 2} = \boldsymbol{Q}^{\boxtimes 2}\left(\boldsymbol{K}^{(l)\boxtimes 2}\right)^\top \tag{A-9}$$

where $\boxtimes$ is the tensor product operator, defined as $[\boldsymbol{A}\boxtimes\boldsymbol{B}]_i = \text{vec}(\boldsymbol{A}_i \otimes \boldsymbol{B}_i)$ (here $\otimes$ is the outer product and vec is vectorization). The low-rank approximation uses a projection matrix $\boldsymbol{P} \in \mathbb{R}^{d_k\times r}$

$$\tilde{\boldsymbol{A}}_{(2)}^{(l)}\boldsymbol{V}^{(l)} = \frac{1}{2}\boldsymbol{Q}^{\boxtimes 2}\boldsymbol{P}\boldsymbol{P}^\top\left(\boldsymbol{K}^{(l)\boxtimes 2}\right)^\top\boldsymbol{V}^{(l)}. \tag{A-10}$$

Applying Johnson–Lindenstrauss Lemma [41] to $\boldsymbol{M} = \boldsymbol{K}^{(l)\boxtimes 2}$, the projection error satisfies

$$\begin{aligned}&\left\|\boldsymbol{Q}^{\boxtimes 2}\left(\left(\boldsymbol{K}^{(l)\boxtimes 2}\right)^\top\boldsymbol{V}^{(l)} - \boldsymbol{P}\boldsymbol{P}^\top\left(\boldsymbol{K}^{(l)\boxtimes 2}\right)^\top\boldsymbol{V}^{(l)}\right)\right\|_F \\ &\le \left\|\boldsymbol{Q}^{\boxtimes 2}\right\|_F\cdot\delta\left\|\boldsymbol{K}^{(l)\boxtimes 2}\right\|_F^2\left\|\boldsymbol{V}^{(l)}\right\|_F.\end{aligned} \tag{A-11}$$

Since $\|\boldsymbol{Q}^{\boxtimes 2}\|_F = \|\boldsymbol{Q}\|_F^2$ and $\|\boldsymbol{K}^{(l)\boxtimes 2}\|_F = \|\boldsymbol{K}^{(l)}\|_F^2$, substituting yields

$$\mathcal{E}_{\text{proj}}^{(2)} \le \delta\cdot\frac{1}{2}\|\boldsymbol{Q}\|_F^2\left\|\boldsymbol{K}^{(l)}\right\|_F^4\left\|\boldsymbol{V}^{(l)}\right\|_F = \delta C\left\|\boldsymbol{V}^{(l)}\right\|_F \tag{A-12}$$

where $C = 1/2\|\boldsymbol{Q}\|_F\|\boldsymbol{K}^{(l)}\|_F^2$. Similar bounds hold for higher order terms ($m \ge 2$) by mathematical induction. For any target error $\epsilon_2 > 0$, to ensure $\mathcal{E}_{\text{proj}} \le \epsilon_2\|\boldsymbol{V}^{(l)}\|_F$, choose $\delta = \epsilon_2/C$ and set the projection dimension $r \ge C'\log(1/\delta)$, where C' is the constant.

C. Total Error Bound

From error decomposition $\mathcal{E}_{\text{total}} \le \mathcal{E}_{\text{Taylor}} + \mathcal{E}_{\text{proj}}$, substituting A-5 and A-12, we have

$$\mathcal{E}_{\text{total}} \le \frac{M}{(k+1)!}\left\|\boldsymbol{T}^{(l)}\right\|_F^{k+1}\left\|\boldsymbol{V}^{(l)}\right\|_F + \delta C\left\|\boldsymbol{V}^{(l)}\right\|_F. \tag{A-13}$$

Setting $\mathcal{E}_{\text{Taylor}} \le (\epsilon/2)\|\boldsymbol{V}^{(l)}\|_F$ and $\mathcal{E}_{\text{proj}} \le (\epsilon/2)\|\boldsymbol{V}^{(l)}\|_F$, we obtain

$$\mathcal{E}_{\text{total}} \le \frac{\epsilon}{2}\left\|\boldsymbol{V}^{(l)}\right\|_F + \frac{\epsilon}{2}\left\|\boldsymbol{V}^{(l)}\right\|_F = \epsilon\left\|\boldsymbol{V}^{(l)}\right\|_F. \tag{A-14}$$

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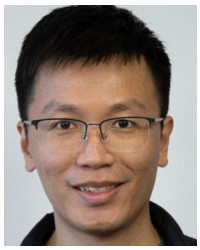
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